9-sinf I BOB MASALALARI YECHIMI

M 1

1. Massasi 2,4 kg bo‘lgan ko‘mirda qancha uglerod atomi borligini hisoblang. Uglerod atomining massasini $2 \cdot 10^{-26}$ kg ga teng deb oling.
2. Hajmi 0,2 mm^3 bo‘lgan moy suv ustida yoyilib, taxminan 0,8 m^2 yuzali yupqa parda hosil qildi. Moy molekulalari suv yuzasida bir qavat bo‘lib tekis yoyilgan deb hisoblab, moy molekulasining chiziqli o‘lchamini aniqlang.
3. Bitta suv molekulasining massasi $3 \cdot 10^{-26}$ kg. Hajmi 5 cm^3 bo‘lgan suvda qancha suv molekulasi bor?
4*. Idishdagi suvda 10^{24} ta suv molekulasi bo‘lsa, suvning hajmi qanday? Suv molekulasining diametrini $3 \cdot 10^{-10}$ m ga teng deb oling.
5*. Moy molekulasining diametri taxminan $2{,}6 \cdot 10^{-10}$ m bo‘lsa, 35 cm^3 moyda qancha moy molekulasi borligini aniqlang.

6. Jadvalni to‘ldiring.

№	Modda	Kimyoviy formulasi	Nisbiy molekulyar massasi (u)
1	Azot		
2	Ozon		
3	Osh tuzi		
4	Metan gazi		
5	Karbonat angidrid		

Berilgan	Formula	Yechilishi
1-mashq		
1. $m = 2.4kg$ $m_0 = 2 * 10^{-26}kg$ $N-?$	$N = \frac{m}{m_0}$	$N = \frac{2.4kg}{2 * 10^{-26}kg} = 1.2 * 10^{26}\ ta$
2. $V = 0.2\ mm^3$ $S = 0.8\ m^2$ $d-?$	$d = \frac{V}{S}$	$d = \frac{0.2 * 10^{-9}\ m^3}{0.8\ m^2} = 0.25 * 10^{-9}m = 0.25\ nm$
3. $m_0 = 3 * 10^{-26}kg$ $V = 5\ sm^3$ $N-?$	$N = \frac{m}{m_0} = \frac{\rho V}{m_0}$	$N = \frac{1000\frac{kg}{m^3} * 5 * 10^{-6}\ m^3}{3 * 10^{-26}kg} = 1.67 * 10^{23}\ ta$
4. $N = 10^{24}$ $d = 3 * 10^{-10}m$ $V-?$	$V = NV_0$ $V_0 = d^3$ $V = Nd^3$	$V = 10^{24} * (3 * 10^{-10}m)^3 = 27 * 10^{-27}\ m^3$
5. $d = 2.6 * 10^{-10}m$ $V = 35\ sm^3$ $N-?$	$N = \frac{V}{V_0}$ $V_0 = d^3$ $N = \frac{V}{d^3}$	$N = \frac{35 * 10^{-6}\ m^3}{(2.6 * 10^{-10}m)^3} = 2 * 10^{24}\ ta$

6. Jadvalni to‘ldiring.

№	Modda	Kimyoviy formulasi	Nisbiy molekulyar massasi (u)
1	Azot	N	*14u*
2	Ozon	O_3	*48 u*
3	Osh tuzi	NaCl	*28 u*
4	Metan gazi	CH_4	*16 u*
5	Karbonat angidrid	CO_2	*44 u*

M 2

1. Massasi 270 g bo‘lgan suvdagi modda miqdorini aniqlang.
2. Miqdori 8 mol bo‘lgan karbonat angidrid (CO_2) gazining massasi nimaga teng?
3. Massasi 7 g bo‘lgan azot (N_2) tarkibidagi molekulalar sonini aniqlang.
4. Moddaning molyar massasi 36 g/molga teng bo‘lsa, shu modda bitta molekulasining massasini aniqlang.
5. Jadvalni to‘ldiring.

№	Moddaning turi	Kimyoviy formulasi	Molyar massasi (g/mol)	Molekulaning massasi (g)
1	Osh tuzi	NaCl		
2	Ozon	O_3		
3	Azot	N_2		
4	Metan gazi	CH_4		

6. Bir dona gaz molekulasining massasi $7{,}33 \cdot 10^{-26}$ kg ga teng. Shu gazning molyar massasini aniqlang.

Berilgan	Formula	Yechilishi
2-mashq		
1. $m = 270\ g$ $M = 18\ g/mol$ $\nu - ?$	$\nu = \frac{m}{M}$	$\nu = \frac{270\ g}{18\ g/mol} = 15\ mol$
2. $\nu = 8\ mol$ $M = 44\ g/mol$ $m - ?$	$m = \nu M$	$m = 8\ mol * 44 \frac{g}{mol} = 352\ g$
3. $m = 7\ g$ $M = 28\ g/mol$ $N - ?$	$N = \frac{m}{M} N_A$	$N = \frac{7\ g}{28\ g/mol} = 6.02 * 10^{23} \frac{1}{mol} = 15 * 10^{22}\ ta$
4. $M = 36\ g/mol$ $m_0 - ?$	$m_0 = \frac{M}{N_A}$	$m_0 = \frac{36\ g/mol}{6.02 * 10^{23} \frac{1}{mol}} = 6 * 10^{-23}\ g$
5 $M_{NaCl} = 22 \frac{g}{mol} + 35 \frac{g}{mol} = 57\ g/mol;$		$m_{0\ NaCl} = \frac{M}{N_A} = \frac{57\ g/mol}{6.02 * 10^{23} \frac{1}{mol}} = 9.5 * 10^{23}\ g$

№	Moddaning turi	Kimyoviy formulasi	Molyar massasi (g/mol)	Molekulaning massasi (g)
1	Osh tuzi	NaCl	*57 g/mol*	*95*□10□^(-22)*
2	Ozon	O_3	*48 g/mol*	*8*□10□^(-23)*
3	Azot	N_2	*28 g/mol*	*4.65*□10□^(-23)*
4	Metan gazi	CH_4	*16 g/mol*	*2.66*□10□^(-23)*

6. $m_0 = 7.33 * 10^{-26} kg$ $M-?$	$M = m_0 N_A$	$M = 7.33 * 10^{-26} kg * 6.02 * 10^{23} \frac{1}{mol} = 44 * 10^{-3}\ kg/mol$

M 3

1. Massasi 81g bo'lgan buyum alyuminiydan yasalgan. Buyumda qancha alyuminiy atomi bor?
2. $4 \cdot 10^{24}$ ta temir atomi bo'lgan jismning massasi qanday?
3. Idishga solingan gazning massasi 5,5 kg ga teng. Idishda $7,5 \cdot 10^{25}$ ta gaz molekulasi mavjud bo'lsa, bu gaz turini aniqlang.
4. Idishga massasi 72 g bo'lgan suv solingan. Idishdagi suv molekulalarining konsentratsiyasini aniqlang.
5. Hajmi 6 cm^3 bo'lgan olmosdagi atomlar sonini aniqlang. Olmosning zichligi 3500 kg/m^3 va molyar massasi 12 g/mol.
6. Modda miqdori 200 mol bo'lgan misdan qalinligi 2 mm bo'lgan tekis mis plastinkasi yasalgan. Plastinkaning yuzasi nimaga teng? Misning zichligi 8900 kg/m^3 va molyar massasi 64 g/mol.
7. Moddaning zichligi 5 g/cm^3 bo'lsa, to'la sirtining yuzi 24 cm^2 bo'lgan kubning massasi qanday bo'ladi?
8. Modda miqdori 34 mol simob qancha hajmni egallaydi? Simobning zichligi 13,6 g/cm^3, molyar massasi 200 g/mol ga teng deb oling.
9. 10 l hajmli idishga 1,6 kg massali kislorod solingan. Idishdagi gaz molekulalarining konsentratsiyasini aniqlang.
10*. Suv molekulasining diametrini $3 \cdot 10^{-10}$ m deb, 3 g suvdagi barcha molekulalar bir-biriga zich qilib bir qatorga joylashtirilsa, qanday uzunlik hosil bo'lishini hisoblang. Bu uzunlikni Yerdan Oygacha bo'lgan masofa ($3,84 \cdot 10^5$ km) bilan taqqoslang.
11*. Idishdagi suvda $3 \cdot 10^{24}$ ta suv molekulasi bo'lsa, suvning hajmi qanday? Suv molekulasining diametri $3 \cdot 10^{-10}$ m ga teng.
12*. Moy molekulasining diametri taxminan $2 \cdot 10^{-10}$ m ga teng bo'lsa, 24 cm^3 moyda qancha moy molekulasi borligini aniqlang.
13. Biror modda molekulasining diametri fotosuratda 0,5 mm ga teng. Agar fotosurat elektron mikroskop yordamida 200 000 marta kattalashtirib olingan bo'lsa, ayni modda molekulasining haqiqiy diametiri qancha?
14. Nima uchun gulxandan chiqayotgan tutun yuqoriga ko'tarilgan sari hatto shamol bo'lmaganda ham ko'zga ko'rinmay ketadi?
15. Nima sababdan singan chinni yoki sopol idishni yelim bilan yopishtirmasa ularni butun holga keltirib bo'lmaydi? Axir jism molekulalari orasida tortishish kuchi mavjud-ku!

Berilgan	Formula	Yechilishi
3-mashq		

1. $m = 81\ g$ $M = 27\ g/mol$ $N-?$	$N = \frac{m}{M}N_A$	$N = \frac{81\ g}{27\ g/mol}6.02 * 10^{23}\frac{1}{mol} = 18.06 * 10^{23}$
2. $N = 4 * 10^{24}$ $M = 127.6\ g/mol$ $m-?$	$m = \nu M$ $\nu = \frac{N}{N_A}$	$m = \frac{4 * 10^{24}}{6.02 * 10^{23}\frac{1}{mol}}127.6\frac{g}{mol} = 84.78\ g$
3. $m = 5.5\ kg$ $N = 7.5 * 10^{25}$ $M-?$	$M = \frac{m}{N}N_A$	$M = \frac{5.5\ kg}{7.5 * 10^{25}}6.02 * 10^{23}\frac{1}{mol} = 4.41 * 10^{-2}\frac{kg}{mol}$ $= 44\ g/mol$
4. $M = 18\ g/mol$ $m = 72\ g$ $n-?$	$n = \frac{N}{V}$ $V = \frac{m}{\rho}$ $m = \frac{N}{N_A}M$ $n = \frac{N_A\rho}{M}$	$n = \frac{6.02 * 10^{23}\frac{1}{mol} * 1000\ kg/m^3}{18 * 10^{-3}\ kg/mol} = 333\ m^{-3}$
5. $\boldsymbol{V = 6\ sm^3}$ $\boldsymbol{\rho = 3500\ kg/m^3}$ $M = 12\ g/mol$ $\boldsymbol{N}-?$	$N = \frac{m}{M}N_A$ $m = \rho V$ $N = \frac{\rho V}{M}N_A$	$N = \frac{\boldsymbol{3500\frac{kg}{m^3} * 6 * 10^{-6}\ m^3}}{12 * 10^{-3}\ kg/mol} = 10^{24}\ ta$
6. $\boldsymbol{\nu = 200\ mol}$ $\boldsymbol{d = 2mm}$ $\boldsymbol{\rho = 8900\ kg/m^3}$ $M = 64 * 10^{-3}\ kg/mol$ $\boldsymbol{S}-?$	$S = \frac{V}{d}$ $V = \frac{m}{\rho}$ $m = \nu M$ $S = \frac{\nu M}{\rho d}$	$S = \frac{\boldsymbol{200\ mol} * 64 * 10^{-3}\frac{kg}{mol}}{\boldsymbol{8900\frac{kg}{m^3}} * 2 * 10^{-3}m} = 0.72\ m^2$
7. $\boldsymbol{S_{to'la} = 24\ sm^2}$ $\boldsymbol{\rho = 5\ g/sm^3}$ $\boldsymbol{m}-?$	$\boldsymbol{S_{to'la} = 24\ sm^2\ bo'lsa\ S = 4sm^2}$ $\boldsymbol{S = 4sm^2\ bo'lsa\ a = \sqrt{S} = \sqrt{4sm^2} = 2\ sm}$ $V = a^3 = (2sm)^3 = 8sm^3$ $m = \rho V = \boldsymbol{5\frac{g}{sm^3}} * 8sm^3 = 40\ g$	
8. $\boldsymbol{\nu = 34\ mol}$ $\boldsymbol{\rho = 13.6\ g/sm^3}$ $M = 200\ g/mol$ $\boldsymbol{V}-?$	$V = \frac{m}{\rho}$ $m = \nu M$ $V = \frac{\nu M}{\rho}$	$V = \frac{\boldsymbol{34\ mol} * \boldsymbol{200}\frac{g}{mol}}{\boldsymbol{13.6\frac{g}{sm^3}}} = 500\ sm^3$
9.V$\boldsymbol{= 10\ l}$ $\boldsymbol{m = 1.6\ kg}$ $M = 32 * 10^{-3}\ kg/mol$ $\boldsymbol{n}-?$	$n = \frac{N}{V}$ $N = \nu N_A$ $\nu = \frac{m}{M}$ $n = \frac{mN_A}{MV}$	$n = \frac{\boldsymbol{1.6\ kg} * 6.02 * 10^{23}\frac{1}{mol}}{32 * 10^{-3}\frac{kg}{mol} * 10 * 10^{-3}m^3} = 3 * 10^{27}m^{-3}$
10. $\boldsymbol{d = 3 * 10^{-10}\ m}$ $\boldsymbol{m = 3\ g}$ $\boldsymbol{R = 3.84 * 10^8 m}$ $M = 18\ g/mol$	$l = dN$ $N = \frac{m}{M}N_A$ $l = d\frac{m}{M}N_A$	$l = \frac{\boldsymbol{3 * 10^{-10}\ m} * \boldsymbol{3 * 10^{-3} kg} * 6.02 * 10^{23}\frac{1}{mol}}{18 * 10^{-3}\frac{kg}{mol}}$ $= 3 * 10^{13}m = 3 * 10^{10}km$

$\frac{l}{R}-$		$\frac{l}{R}=\frac{3*10^{10}km}{\mathbf{3.84*10^5km}}=78000$
11. $\boldsymbol{N=3*10^{24}}$ $\boldsymbol{d=3*10^{-10}m}$ $\boldsymbol{V}-?$	$V=NV_0$ $V_0=d^3$ $V=Nd^3$	$V=\mathbf{3*10^{24}}*(\mathbf{3*10^{-10}m})^3=\mathbf{81*10^{-6}m^3}$ $=\mathbf{81sm^3}$
12.$\boldsymbol{d=2*10^{-10}\,m}$ $\boldsymbol{V=24\,sm^3}$ $\boldsymbol{N}-?$	$N=\frac{V}{V_0}$ $V_0=d^3$ $N=\frac{V}{d^3}$	$N=\frac{24*10^{-6}\,m^3}{(\mathbf{2*10^{-10}\,m})^3}=3*10^{24}$
13. $\boldsymbol{d=0.5\,mm}$ $\boldsymbol{K=200000}$ $\boldsymbol{d_0}-?$	$d_0=\frac{d}{K}$	$d_0=\frac{0.5*10^{-3}m}{2*10^5}=0.25*10^{-8}m=2.5*10^{-9}m$
14.Konsentratsiya kamayib ketadi		
15. G'adir-budirlik mavjud		

M 4

1. Idish vodorod gazi bilan to'ldirilgan. Idishdagi gaz molekulalarining konsentratsiyasi $4{,}5\cdot10^{24}$ m^{-3} ga teng. Idishdagi gaz bosimini hisoblang. Gaz molekulalarining o'rtacha kvadratik tezligini 400 m/s ga teng deb oling.
2. Ideal gaz molekulalarining o'rtacha kvadratik tezligi 600 m/s va zichligi 0,9 kg/m^3 bo'lsa, idishdagi gaz bosimini aniqlang.
3. Idishdagi gazning zichligi 1,5 kg/m^3 va bosimi 7,2 kPa bo'lgan gaz molekulalarining o'rtacha kvadratik tezligi nimaga teng?
4. Idishdagi hajm birligidagi molekulalar soni $3\cdot10^{25}$ m^{-3} va bosimi 80 kPa bo'lgan gaz molekulalarining o'rtacha kinetik energiyasini hisoblang.
5. Idishga solingan kislorod gazi, idish devoriga 90 kPa bosim bermoqda. Agar kislorod molekulalari 600 m/s o'rtacha kvadratik tezlik bilan harakatlanayotgan bo'lsa, idishdagi gaz zichligi qanday bo'lgan?
6. Massasi 0,3 kg bo'lgan gaz 400 kPa bosimda 1 m^3 hajmni egallasa, uning molekulalari harakatining o'rtacha kvadratik tezliklari qanday bo'ladi?
7. 30 kPa bosimda bir atomli gaz molekulasining o'rtacha kinetik energiyasini toping. Berilgan bosimda bu gaz molekulalarining konsentratsiyasi $4\cdot10^{25}$ m^{-3} ga teng.

Berilgan	Formula	Yechilishi
	4-mashq	
1. n= $4.5*10^{24}m^{-3}$ $v=400\,m/s$ $p-?$	$p=\frac{nm_0v^2}{3}$ $m_0=\frac{M}{N_A}$	$[p]=\frac{m^{-3}kg\frac{m^2}{s^2}}{1}=\frac{kg}{ms^2}=\frac{kgN}{mkgm}=\frac{N}{m^2}=Pa$ $m_0=\frac{2*10^{-3}\,kg/mol}{6.02*10^{23}mol^{-1}}=0.34*10^{-26}kg$ $p=\frac{4.5*10^{24}m^{-3}*0.34*10^{-26}kg*(400\,m/s)^2}{3}$ $=800\,Pa$
2. $v=600\frac{m}{s}$ $\rho=0.9\,kg/m^3$ $p-?$	$p=\frac{1}{3}\rho v^2$	$[p]=\frac{kg}{m^3}*\frac{m^2}{s^2}=\frac{kg*m}{s^2}*\frac{1}{m^2}=\frac{N}{m^2}=Pa$

		$p=\frac{1}{3}0.9(600)^2=108000Pa=108\ kPa$
3.$\rho=1.5\ kg/m^3$ $p=7.2\ kPa$ v-?	$v=\sqrt{\frac{3p}{\rho}}$	$[v]=\sqrt{\frac{N/m^2}{kg/m^3}}=\sqrt{\frac{N*m}{kg}}=\sqrt{\frac{kg*\frac{m}{s^2}*m}{kg}}=\frac{m}{s}$ $v=\sqrt{\frac{3*7.2*10^3}{1.5}}=\sqrt{144*10^2}\frac{m}{s}=120\ m/s$
4. $n=3*10^{25}m^{-3}$ $p=80\ kPa$ $E_K-?$	$p=\frac{2}{3}nE_K$ $E_K=\frac{3p}{2n}$	$[E]=\frac{\frac{N}{m^2}}{\frac{1}{m^3}}=N*m=J$ $E_K=\frac{3*80*10^3}{3*10^{25}}J=4*10^{-21}J$
5. $p=7.2\ kPa$ $v=600\frac{m}{s}$ $\rho-?$	$p=\frac{1}{3}\rho v^2$ $\rho=\frac{3p}{v^2}$	$[\rho]=\frac{\frac{N}{m^2}}{\frac{m^2}{s^2}}=\frac{\frac{kg*m/s^2}{m^2}}{\frac{m^2}{s^2}}=kg/m^3$ $\rho=\frac{3*7.2*10^3}{(600)^2}=0.75\ kg/m^3$
6. $\boldsymbol{m=0.3\ kg}$ $\boldsymbol{p=400\ kPa}$ $\boldsymbol{V=1m^3}$ $\boldsymbol{v}-?$	$v=\sqrt{\frac{3p}{\rho}}$ $\rho=\frac{m}{V}$ $v=\sqrt{\frac{3pV}{m}}$	$[v]=\sqrt{\frac{\frac{N}{m^2}m^3}{kg}}=\sqrt{\frac{kg*\frac{m}{s^2}}{\frac{m^2}{kg}m^3}}=\sqrt{\frac{m^2}{s^2}}=m/s$ $v=\sqrt{\frac{3*4*10^5*1}{0.3}}\frac{m}{s}=\frac{\sqrt{4*10^6}m}{s}=2000m/s$
7. $\boldsymbol{p=30\ kPa}$ $\boldsymbol{n=4*10^{25}m^{-3}}$ $\boldsymbol{E_K}-?$	$E_K=\frac{3p}{2n}$	$[E]=\frac{\frac{N}{m^2}}{\frac{1}{m^3}}=N*m=J$ $E_K=\frac{3*30*10^3}{4*10^{25}}J=1.125*10^{-21}J$

M 5

1. Quyidagi Selsiy shkalasida ifodalangan temperaturalarni Kelvin shkalasida ifodalang: 0 °C, 27 °C, 100 °C, 127 °C, -73 °C, -223 °C, -200 °C.
2. Quyidagi Kelvin shkalalarida ifodalangan temperaturalarni Selsiy shkalalarida ifodalang: 0 K, 73 K, 273 K, 300 K, 773 K, 1000 K, 2000 K.
3. Yopiq idishdagi gaz 27 °C dan 627 °C gacha qizdirildi. Bunda gaz molekulalarining idish devoriga beradigan bosimi qanday o'zgaradi?
4. Idishdagi gaz molekulalarining konsentratsiyasi $3\cdot10^{27}m^{-3}$ga teng. Idish ichidagi temperatura 60 °C bo'lganda gaz molekulalarining idish devoriga beradigan bosimi qanday bo'ladi?
5. Idish ichidagi gazning temperaturasi 400 K bo'lganda, manometr idishdagi gaz bosimi 276 kPa ga teng bo'lganligini ko'rsatdi. Idishdagi gaz molekulalarining konsentratsiyasi nimaga teng?
6. Normal sharoitda 1 m^3 hajmdagi havo molekulalarining sonini baholang. Normal sharoit uchun bosimni 100 kPa, temperaturani 273 K ga teng deb oling.
7. Maxsus so'ruvchi nasos yordamida idishdan havo so'rilib, uning ichida 1 pPa bosimdagi vakuum hosil qilindi. Vakuumning 1 cm^3 hajmida qancha gaz molekulasi bor? Idish ichidagi temperatura 300 K.

Berilgan	Formula	Yechilishi
5-mashq		

1.

t	Formula	T
0	T=t+273	273
27	T=t+273	300
100	T=t+273	373
127	T=t+273	400
-73	T=t+273	200
-223	T=t+273	50
-200	T=t+273	73

2.

T	Formula	t
0	t=T-273	-273
73	t=T-273	-200
273	t=T-273	0
300	t=T-273	27
773	t=T-273	500
1000	t=T-273	727
2000	t=T-273	1727

3. $t_1 = 27\ ℃$ $t_2 = 627\ ℃$ $\frac{p_2}{p_1} - ?$	$\frac{p_2}{p_1} = \frac{nkT_2}{nkT_1}$	$\frac{p_2}{p_1} = \frac{900K}{300K} = 3$
4. $n = 3 * 10^{27} m^{-3}$ $t = 60℃$ $p - ?$	$p = nkT$ $T = t + 273$	$[p] = \frac{1}{m^3} * \frac{J}{K} * K = \frac{N}{m^2} = Pa$ $p = 3 * 10^{27} * 1.38 * 10^{-23} * 333Pa = 1378{,}62 * 10^4 Pa = 13786.2\ kPa$=13.8 MPa **(Darslikdagi javobi noto'g'ri)**
5. $T = 400\ K$ $p = 276\ kPa$ $n - ?$	$n = \frac{p}{kT}$	$[n] = \frac{Pa}{\frac{J}{K} * K} = \frac{\frac{N}{m^2}}{N * m} = \frac{1}{m^3}$ $n = \frac{276 * 10^3}{1.38 * 10^{-23} * 400} m^{-3} = 0.5 * 10^{26} m^{-3}$
6. $\boldsymbol{V = 1\ m^3}$ $\boldsymbol{p = 100\ kPa}$ $\boldsymbol{T = 273\ K}$ $\boldsymbol{N} - ?$	$N = nV$ $n = \frac{p}{kT}$ $N = \frac{pV}{kT}$	$[N] = \frac{\frac{N}{m^2} * m^3}{\frac{J}{K} * K} = \frac{\frac{N}{m^2} * m^3}{N * m} = 1$ $N = \frac{10^5 * 1}{1.38 * 10^{-23} * 273} = 26 * 10^{24}$
7. $\boldsymbol{p = 1\ pPa}$ $\boldsymbol{V = 1sm^3}$ $\boldsymbol{T = 300K}$ $\boldsymbol{N} - ?$	$N = \frac{pV}{kT}$	$[N] = \frac{\frac{N}{m^2} * m^3}{\frac{J}{K} * K} = \frac{\frac{N}{m^2} * m^3}{N * m} = 1$ $N = \frac{10^{-12} * 10^{-6}}{1.38 * 10^{-23} * 273} = 241$

M 6

1. Vodorod molekulasining –23 °C temperaturadagi o‘rtacha kvadratik tezligini aniqlang.
2. Qanday temperaturada kislorod molekulasining o‘rtacha kvadratik tezligi 500 m/s ga teng bo‘ladi?
3. Gaz molekulalari ilgarilanma harakatining o‘rtacha kinetik energiyasi $9{,}52\cdot10^{-21}$ J bo‘lgan gazning temperaturasini aniqlang.
4. Molekulalar konsentratsiyasi $4\cdot10^{26}$ m^{-3} va bosimi $1{,}6\cdot10^{5}$ Pa bo‘lgan bir atomli gaz molekulalarining o‘rtacha kinetik energiyasi nimaga teng?
5. Bir atomli gaz molekulalarining ilgarilanma harakati o‘rtacha kinetik energiyasi $1{,}2\cdot10^{-20}$ J va bosimi 2,4 MPa bo‘lsa, shu gaz molekulalarining konsentratsiyasi qanday bo‘ladi?
6. Miqdori ikki mol bo‘lgan gazning idish devorlariga beradigan bosimi 10 kPa ga teng. Gaz egallab turgan hajmni aniqlang. Gazning temperaturasi 300 K.
7*. Qanday temperaturadagi geliy molekulalarining o‘rtacha kvadratik tezligi, 350 K temperaturadagi vodorod molekulalarining o‘rtacha kvadratik tezligiga teng bo‘ladi?
8*. Gaz temperaturasi 150 °C ga oshirilganda, molekulalarining o‘rtacha kvadratik tezligi 300 m/s dan 600 m/s gacha ortdi. Gazning dastlabki temperaturasi qanday bo‘lgan?

Berilgan	Formula	Yechilishi
6-mashq		
1. $M = 2*10^{-3}\frac{kg}{mol}$ $t = -23℃$ $v-?$	$v = \sqrt{\frac{3RT}{M}}$ $T = t + 273$	$v = \sqrt{\frac{\frac{J}{mol*K}K}{\frac{kg}{mol}}} = \sqrt{\frac{N*m}{kg}} = \sqrt{\frac{kg*\frac{m}{s^2}*m}{kg}} = m/s$ $v = \sqrt{\frac{3*8.31*250}{2*10^{-3}}\frac{m}{s}} = \sqrt{312*10^4}\frac{m}{s} = 1765\ m/s$
2. $v = 500\frac{m}{s}$ $M = 32*10^{-3}\frac{kg}{mol}$ $T-?$	$v = \sqrt{\frac{3RT}{M}}$ $v^2 = \frac{3RT}{M}$ $T = \frac{v^2M}{3R}$	$T = \frac{(\frac{m}{s})^2\frac{kg}{mol}}{\frac{J}{mol*K}} = \frac{(\frac{m}{s})^2\frac{kg}{mol}}{\frac{N*m}{mol*K}} = \frac{(\frac{m}{s})^2\frac{kg}{mol}}{\frac{kg*\frac{m}{s^2}*m}{mol*K}} = K$ $T = \frac{(500)^2*32*10^{-3}}{3*8.31}K = 320K$
3. $E = 9.52*10^{-21}J$ $T-?$	$E_K = \frac{3}{2}kT$ $T = \frac{2E_K}{3k}$	$[T] = \frac{J}{\frac{J}{K}} = K$ $T = \frac{2*9.52*10^{-21}}{3*1.38*10^{-23}}K = 460\ K$
4. $n = 4*10^{26}m^{-3}$ $p = 1.6*10^5Pa$ $E_K-?$	$E_K = \frac{3p}{2n}$	$[E] = \frac{Pa}{m^{-3}} = \frac{\frac{N}{m^2}}{m^{-3}} = \frac{N}{m^2}*m^3 = J$ $E_K = \frac{3*1.6*10^5}{2*4*10^{26}}J = 6*10^{-22}J$

5. $E_K = 1.2 * 10^{-20} J$ $p = 2.4\ MPa$ $n-?$	$n = \frac{3p}{2E_K}$	$[n] = \frac{Pa}{J} = \frac{\frac{N}{m^2}}{N * m} = \frac{1}{m^3} = m^{-3}$ $n = \frac{3 * 2.4 * 10^6}{2 * 1.2 * 10^{-20}} m^{-3} = 3 * 10^{26} m^{-3}$
6. $\boldsymbol{\nu = 2\ mol}$ $\boldsymbol{p = 10\ kPa}$ $\boldsymbol{T = 300\ K}$ $\boldsymbol{V}-?$	$p = nkT$ $n = \frac{N}{V}$ $N = \nu N_A$ $V = \frac{\nu N_A kT}{p}$	$p = nkT = \frac{N}{V} kT = \frac{\nu N_A}{V} kT \qquad V = \frac{\nu N_A kT}{p}$ $[V] = \frac{mol \frac{1}{mol} \frac{J}{K} K}{\frac{N}{m^2}} = \frac{mol \frac{1}{mol} \frac{N * m}{K} K}{\frac{N}{m^2}} = m^3$ $V = \frac{2 * 6.02 * 10^{23} * 1.38 * 10^{-23} * 300}{10^4} m^3 = 0.5 m^3$
7. $M_1 = 4 * 10^{-3} \frac{kg}{mol}$ $T_2 = 350K$ $M_2 = 2 * 10^{-3} \frac{kg}{mol}$ $v_1 = v_2$ $T_1-?$	$T_1 = \frac{T_2 M_1}{M_2}$	$v_1 = \sqrt{\frac{3RT_1}{M_1}} \quad v_2 = \sqrt{\frac{3RT_2}{M_2}} \quad \frac{3RT_1}{M_1} = \frac{3RT_2}{M_2} \quad T_1 = \frac{T_2 M_1}{M_2}$ $[T] = \frac{K * \frac{kg}{mol}}{\frac{kg}{mol}} = K$ $T_1 = \frac{350 * 4 * 10^{-3}}{2 * 10^{-3}} K = 700\ K$
8. $\boldsymbol{\Delta t = 150}$℃ $v_1 = 300 \frac{m}{s}$ $v_2 = 600 \frac{m}{s}$ $T_1-?$	$T_1 = \frac{v_1^2 \Delta T}{v_2^2 - v_1^2}$ $[T] = \frac{\frac{m^2}{s^2} K}{\frac{m^2}{s^2}}$=K	$\Delta T = T_2 - T_1$ $T_2 = T_1 + \Delta T$ $v_1 = \sqrt{\frac{3RT_1}{M}}$ $v_2 = \sqrt{\frac{3RT_2}{M}} = \sqrt{\frac{3R(T_1 + \Delta T)}{M_1}}$ $\frac{v_2}{v_1} = \frac{\sqrt{\frac{3R(T_1 + \Delta T)}{M_1}}}{\sqrt{\frac{3RT_1}{M}}} = \sqrt{\frac{T_1 + \Delta T}{T_1}}$ $\frac{v_2^2}{v_1^2} = \frac{T_1 + \Delta T}{T_1}$ $v_2^2 * T_1 = v_1^2 T_1 + v_1^2 \Delta T$ $T_1(v_2^2 - v_1^2) = v_1^2 \Delta T$ $T_1 = \frac{v_1^2 \Delta T}{v_2^2 - v_1^2}$ $T_1 = \frac{(300\ m/s)^2 * 150}{(600\ m/s)^2 - (300\ m/s)^2} K = 50K$

M 7

1. Bosimi 0,45 MPa va temperaturasi 52°C bo'lganda 500 mol gaz qanday hajmni egallaydi?
2. Hajmi 0,05 m^3 va temperaturasi 500 K bo'lgan gazning bosimi 250 kPa. Modda miqdorini aniqlang.
3. Massasi 8 g bo'lgan gaz 27 °C temperaturada va 150 kPa bosimda 4,15 *l* hajmni egallaydi. Bu qanday gaz?
4. Temperaturasi 367 °C va bosimi 415 kPa bo'lgan kislorod gazining zichligi qanday?
5. 24 *l* hajmli ballonda 1,2 kg karbonat angidrid gazi bor. Ballon $3 \cdot 10^6$ Pa bosimgacha chidaydi. Qanday temperaturada portlash xavfi tug'iladi?
6. Hajmi 40 *l* bo'lgan idishga gaz solingan bo'lib, uning temperaturasi 400 K va bosimi 200 kPa ga teng. Idishdagi gazning miqdorini aniqlang.
7. Temperaturasi 17 °C bo'lgan 4x5x3 m^3 o'lchamli xonadagi havo miqdorini aniqlang. Atmosfera bosimi 10^5 Pa ga teng deb oling.
8. Hajm 16,6 *l* bo'lgan idishda 280 g azot gazi 3,5 MPa bosim ostida bo'lsa, uning temperaturasi nimaga teng?

Berilgan	Formula	Yechilishi
7-mashq		
$p = 0.45\ MPa$ $t = 52℃$ $\nu = 500\ mol$ $V-?$	$pV = \nu RT$ $V = \frac{\nu RT}{p}$ $T = t + 273$	$[V] = \frac{mol * \frac{J}{mol*K}K}{\frac{N}{m^2}} = \frac{N*m}{\frac{N}{m^2}} = m^3$ $V = \frac{500 * 8.31 * 325}{0.45 * 10^6} m^3 = 3m^3$
2. $V = 0.05m^3$ $T = 500K$ $p = 250\ kPa$ $\nu-?$	$pV = \nu RT$ $\nu = \frac{pV}{RT}$	$\nu = \frac{\frac{N}{m^2} * m^3}{\frac{J}{mol*K}K} = \frac{\frac{N}{m^2} * m^3}{\frac{N*m}{mol*K}K} = mol$ $\nu = \frac{250 * 10^3 * 0.05}{8.31 * 500} mol = 3mol$
3. $m = 8g$ $t = 27℃$ $p = 150\ kPa$ $V = 4.15\ l$ $M-?$	$pV = \frac{m}{M}RT$ $M = \frac{mRT}{pV}$	$[M] = \frac{kg * \frac{J}{mol*K}K}{\frac{N}{m^2}m^3} = \frac{kg * \frac{N*m}{mol*K}K}{\frac{N}{m^2}m^3} = kg/mol$ $M = \frac{8 * 10^{-3} * 8.31 * 300}{150 * 10^3 * 4.15 * 10^{-3}} \frac{kg}{mol} = 32 * 10^{-3}\ kg/mol$
4. $\boldsymbol{t = 367℃}$ $\boldsymbol{p = 415\ kPa}$ $\boldsymbol{M = 32 * 10^{-3}\ \frac{kg}{mol}}$ $\boldsymbol{\rho}-?$	$\rho = \frac{m}{V}$ $m = \frac{pVM}{RT}$ $\rho = \frac{pM}{RT}$	$[\rho] = \frac{Pa * \frac{kg}{mol}}{\frac{J}{mol*K}K} = \frac{\frac{N}{m^2} * \frac{kg}{mol}}{\frac{N*m}{mol*K}K} = kg/m^3$ $T = t + 273 = 640K$ $\rho = \frac{415 * 10^3 * 32 * 10^{-3}}{8.31 * 640} \frac{kg}{m^3} = 2.5\ kg/m^3$
5. $\boldsymbol{V = 24\ l = 24 * 10^{-3} m^3}$ $\boldsymbol{m = 1.2\ kg}$ $\boldsymbol{M = 44 * 10^{-3}\ \frac{kg}{mol}}$ $\boldsymbol{p = 3 * 10^6\ Pa}$ $\boldsymbol{T}-?$	$pV = \frac{m}{M}RT$ $T = \frac{pVM}{mR}$	$[T] = \frac{Pa * m^3 * \frac{kg}{mol}}{kg * \frac{J}{mol*K}} = \frac{\frac{N}{m^2} * m^3 * \frac{kg}{mol}}{kg * \frac{N*m}{molK}} = K$ $T = \frac{3 * 10^6 * 24 * 10^{-3} * 44 * 10^{-3}}{1.2 * 8.31} = 317.7\ K$

6. $\boldsymbol{V = 40\ l = 40 * 10^{-3} m^3}$ $\boldsymbol{T = 400\ K}$ $\boldsymbol{p = 200\ kPa}$ $\boldsymbol{\nu}-?$	$\nu = \frac{PV}{RT}$	$[\nu] = \frac{\frac{N}{m^2} * m^3}{N * \frac{m}{mol * K} K} = mol$ $\nu = \frac{2 * 10^5 * 40 * 10^{-3}}{8.31 * 400} mol = 2.4\ mol$
7. $\boldsymbol{t = 17}$℃ $\boldsymbol{V = (4 * 5 * 3) m^3}$ $\boldsymbol{p = 10^5 Pa}$ $\boldsymbol{\nu}-?$	$\nu = \frac{PV}{RT}$	$[\nu] = \frac{\frac{N}{m^2} * m^3}{N * \frac{m}{mol * K} K} = mol$ $\nu = \frac{10^5 * 60 * 10^{-3}}{8.31 * 290} mol = 2500\ mol$
8. $\boldsymbol{V = 16.6\ l}$ $\boldsymbol{m = 280 g}$ $\boldsymbol{p = 3.5\ MPa}$ $\boldsymbol{M = 28 * 10^{-3} \frac{kg}{mol}}$ $\boldsymbol{T}-?$	$pV = \frac{m}{M} RT$ $T = \frac{pVM}{mR}$	$[T] = \frac{Pa * m^3 * \frac{kg}{mol}}{kg * \frac{J}{mol * K}} = \frac{\frac{N}{m^2} * m^3 * \frac{kg}{mol}}{kg * \frac{N * m}{molK}} = K$ $T = \frac{3.5 * 10^6 * 16.6 * 10^{-3} * 28 * 10^{-3}}{0.28 * 8.31} = 700\ K$

M 8

1. Gazning dastlabki hajmi 0,2 *l*, bosimi esa 300 kPa bo'lgan. Gaz izotermik kengayib, bosimi 120 kPa ga erishdi. Gazning keyingi hajmini toping.

2. Porshenli silindr ichiga qamalgan gazning dastlabki hajmi 24 cm³, bosimi 0,8 MPa bo'lgan. Gaz izotermik siqilib, gazning hajmi 16 cm³ ga keltirilganda uning bosimi qanday qiymatga erishadi?

3. Normal atmosfera bosimi sharoitida ideal gaz 50 *l* hajmni egallaydi. Agar bosim 4 marta ortsa, gaz qancha hajmni egallaydi? Temperatura o'zgarmas.

4. Ideal gaz 1,2 *l* hajmdan 0,8 *l* hajmgacha izotermik siqildi. Bunda gazning bosimi 40 kPa ga ortdi. Gazning dastlabki bosimi qanday bo'lgan?

Berilgan	Formula	Yechilishi
	8-mashq	
1. $V_1 = 0.2\ l$ $p_1 = 300\ kPa$ $T = const$ $p_2 = 120\ kPa$ $V_2-?$	$p_1V_1 = p_2V_2$ $V_2 = \frac{p_1V_1}{p_2}$	$[V] = \frac{Pa * m^3}{Pa} = m^3$ $V_2 = \frac{3 * 10^5 * 0.2 * 10^{-3}}{1.2 * 10^5} m^3 = 0.5 * 10^{-3}\ m^3$
2. $V_1 = 24\ sm^3$ $p_1 = 0.8\ MPa$ $T = const$ $V_2 = 16\ sm^3$ $p_2-?$	$p_1V_1 = p_2V_2$ $p_2 = \frac{p_1V_1}{V_2}$	$[Pa] = \frac{Pa * m^3}{m^3} = Pa$ $p_2 = \frac{0.8 * 10^6 * 24 * 10^{-6}}{16 * 10^{-6}} Pa = 1.2 * 10^6\ Pa\ 1.2\ MPa$
3. $V_1 = 50\ l$ $p_2 = 4p_1$ $T = const$ $V_2-?$	$p_1V_1 = p_2V_2$ $V_2 = \frac{p_1V_1}{4p_1}$ $V_2 = \frac{V_1}{4}$	$[V] = \frac{Pa * m^3}{Pa} = m^3$ $V_2 = \frac{50 * 10^{-3}}{4} m^3 = 12.5 * 10^{-3}\ m^3 = 12.5\ l$
4. $V_1 = 1.2\ l$	$\Delta p = p_2 - p_1$	$\Delta p = p_2 - p_1$ $\quad p_2 = \frac{p_1V_1}{V_2}$

$V_2 = 0.8\ l$ $T = const$ $\Delta p = 40\ kPa$ $p_1 - ?$	$p_2 = \frac{p_1 V_1}{V_2}$ $p_1 = \frac{\Delta p V_2}{V_1 - V_2}$ $\Delta T = T_2 - T_1$	$\Delta p = p_2 - p_1 = \frac{p_1 V_1}{V_2} - p_1 = \frac{p_1 V_1 - p_1 V_2}{V_2}$ $\Delta p = \frac{p_1 V_1 - p_1 V_2}{V_2}$ $p_1 = \frac{\Delta p V_2}{V_1 - V_2}$ $[p] = \frac{Pa * m^3}{m^3} = Pa$ $p_1 = \frac{4 * 10^4 * 0.8}{1.2 - 0.8} Pa = 8 * 10^4\ Pa = 80\ kPa$

M 9

1. Temperaturasi 27 °C bo‘lgan ideal gazning hajmi 10 *l* edi. Gaz izobarik ravishda 327 °C gacha isitilgandagi hajmi qanday o‘zgaradi?
2. Ideal gazning temperaturasi 51 °C va hajmi 0,9 *l* . Bosim o‘zgarmaganda, hajm 0,3 *l* ga teng bo‘lishi uchun gazni qancha sovitish kerak?
3. Gaz 27 °C temperaturada 3 *l* hajmga ega. Bu gaz izobarik 100 °C da qizdirilsa, u qanday hajmni egallaydi?
4*. Ideal gaz 47 °C da 3 *l* hajmni egallagan. Bosimni o‘zgartirmasdan, hajmni 1,2 *l* ga orttirish uchun gazning temperaturasini qanchaga ko‘tarish kerak?

Berilgan	Formula	Yechilishi
	9-mashq	
1. $t_1 = 27$ °C $V_1 = 10\ l$ $p = const$ $t_2 = 327$ °C $\Delta V - ?$	$\frac{V_1}{V_2} = \frac{T_1}{T_2}$ $V_2 = \frac{V_1 T_2}{T_1}$ $\Delta V = V_2 - V_1$ $T = t + 273$	$[V] = \frac{m^3 * K}{K} = m^3$ $V_2 = \frac{10 * 10^{-3} * 600}{300} m^3 = 20 * 10^{-3}\ m^3 = 20\ l$ $\Delta V = 20\ l - 10\ l = 10\ l$
2. $t_1 = 51$ °C $V_1 = 0.9\ l$ $p = const$ $V_2 = 0.3\ l$ $t_2 - ?$ $\Delta t - ?$	$T = t + 273$ $\frac{V_1}{V_2} = \frac{T_1}{T_2}$ $T_2 = \frac{V_2 T_1}{V_1}$	$[T] = \frac{m^3 * K}{m^3} = K$ $T_2 = \frac{0.3 * 10^{-3} * 324}{0.9 * 10^{-3}} K = 108K$ $\Delta T = 108 - 324 = -216K$
3. $t_1 = 27$ °C $V_1 = 3\ l$ $p = const$ $\Delta t = 100$°C $V_2 - ?$	$t_2 = \Delta t + t_1$ $V_2 = \frac{V_1 T_2}{T_1}$	$[V] = \frac{m^3 * K}{K} = m^3$ $V_2 = \frac{3 * 10^{-3} * 400}{300} m^3 = 4 * 10^{-3}\ m^3 = 4\ l$
4. $t_1 = 47$ °C $V_1 = 3\ l$ $p = const$ $V_2 = V_1 + 1.2l = 4.2l$ $T_2 - ?\ \Delta T - ?$	$T_2 = \frac{V_2 T_1}{V_1}$ $T = t + 273$ $\Delta T = T_2 - T_1$	$[T] = \frac{m^3 * K}{m^3} = K$ $T_2 = \frac{4.2 * 320}{3\ l} K = 448K$ $\Delta T = (448 - 320)K = 128K$

M 10

1. Ballondagi gaz 17 °C temperaturada $1{,}45 \cdot 10^5$ Pa bosimga ega bo‘lsa, qanday temperaturada uning bosimi $2 \cdot 10^5$ Pa bo‘ladi?
2. Agar cho‘g‘lanma lampochka yonganda, temperaturasi 17 °C dan 360 °C gacha ko‘tarilsa, uning ichidagi gaz bosimi qanday o‘zgaradi?
3. Gaz 300 K dan 420 K gacha izoxorik qizdirilganda uning bosimi 50 kPa ga ortdi. Gaz dastlab qanday bosimda bo‘lgan?

Berilgan	Formula	Yechilishi
10-mashq		
1. $t_1 = 17℃$ $p_1 = 1.45 * 10^5 Pa$ $p_2 = 2 * 10^5 Pa$ $V = const$ $t_2 - ?$	$\frac{p_1}{p_2} = \frac{T_1}{T_2}$ $T_2 = \frac{T_1 p_2}{p_1}$ $T = t + 273$	$[T] = \frac{K * Pa}{Pa} = K$ $T_2 = \frac{290 * 2 * 10^5}{1.45 * 10^5} K = 400K$
2. $t_1 = 17℃$ $t_2 = 360℃$ $\frac{p_2}{p_1} - ?$	$\frac{p_1}{p_2} = \frac{T_1}{T_2}$ $\frac{p_2}{p_1} = \frac{T_2}{T_1}$	$\frac{p_2}{p_1} = \frac{633}{290} = 2.2$
3. $T_1 = 300K$ $T_2 = 420K$ $\Delta p = 50\ kPa$ $P_1 - ?$	$p_1 = \frac{\Delta p T_1}{T_2 - T_1}$	$\Delta p = p_2 - p_1$ $\quad p_2 = \Delta p + p_1$ $\frac{p_1}{p_2} = \frac{T_1}{T_2}$ $\quad p_1 = \frac{p_2 T_1}{T_2} = \frac{(\Delta p + p_1) T_1}{T_2}$ $p_1 T_2 = (\Delta p + p_1) T_1$ $p_1 = \frac{\Delta p T_1}{T_2 - T_1}$ $[p] = \frac{PaK}{K} = K$ $p_1 = \frac{50 * 10^3 * 300}{420 - 300} Pa = 125 * 10^3 Pa = 125\ kPa$

1. 27 °C temperaturada yopiq idishdagi gazning bosimi 900 kPa ga teng bo‘lgan. Gaz qizdirilib, temperaturasi 227 °C ga yetkazilganda idish ichidagi gazning bosimi qanday qiymatga erishadi?

Berilgan	Formula	Yechilishi
$t_1 = 27℃$ $p_1 = 900\ kPa$ $t_1 = 227℃$ $p_2 - ?$ $V = const$	$\frac{p_1}{p_2} = \frac{T_1}{T_2}$ $p_2 = \frac{p_1 T_2}{T_1}$	$[p] = \frac{Pa * K}{K} = Pa$ $p_2 = \frac{900 * 10^3 * 500}{300} Pa = 1500 * 10^3 Pa = 1500\ kPa$

2. Ballonda 17 °C temperatura gaz bor. Agar gazning 0,4 qismi chiqib ketsa va bunda temperatura 10 °C ga pasaysa, ballondagi gazning bosimi qanday o‘zgaradi?

Berilgan	Formula	Yechilishi
$t_1 = 17℃$ $t_2 = t_1 - 10℃$ $m_2 = m_1 - 0.4m_1$ $\frac{p_2}{p_1} - ?$	$\frac{p_2}{p_1} = \frac{m_2 T_2}{m_1 T_1}$	$p_1 V_1 = \frac{m_1}{M} R T_1$ $\quad p_2 V_2 = \frac{m_2}{M} R T_2$ $V_1 = \frac{m_1}{M p_1} R T_1$ $\quad V_2 = \frac{m_2}{M p_2} R T_2$ $V_1 = V_2$ $\frac{m_1}{M p_1} R T_1 = \frac{m_2}{M p_2} R T_2$ $\frac{m_1}{p_1} T_1 = \frac{m_2}{p_2} T_2$ $\frac{m_1}{p_1} T_1 = \frac{0.6 m_1}{p_2} T_2$ $\frac{p_2}{p_1} = \frac{0.6 m_1 T_2}{m_1 T_1} = \frac{0.6 * 7}{17} = 0.25$

3. Dastlabki temperaturasi 27 °C bo‘lgan ideal gaz izobarik kengayib, uning hajmi 24 % ga ortdi. Uning keyingi temperaturasi qanday bo‘lgan?

Berilgan	Formula	Yechilishi
$t_1 = 27℃$ $P = const$ $V_2 = 1.24V_1$ $t_2 - ?$	$T_2 = \frac{T_1 V_2}{V_1}$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}\ dan\ T_2 = \frac{T_1 V_2}{V_1}$ $T_2 = \frac{T_1 1.24 V_1}{V_1} = 1.24T_1 = 1.24 * 300K = 372K = 99℃$ $[T] = \frac{K * m^3}{m^3} = K$

4. Ideal gaz o‘zgarmas bosimda 27 °C dan 117 °C gacha isitilganda, gaz hajmi necha foizga ortadi?

Berilgan	Formula	Yechilishi
$t_1 = 27℃$ $t_2 = 117℃$ $V_2 = xV_1$	$\frac{V_2}{V_1} = \frac{T_2}{T_1}$	$\frac{V_2}{V_1} = \frac{390}{300} = 1.3$ $V_2 = 1.3V_1$ Gaz hajmi 1.3 baravar, ya’ni 30% ortgan

5. Havo putakchasi suv havzasining tubidan suv yuziga chiqquncha 3,5 marta kattalashdi. Suv havzasining chuqurligi qanday? Suvning ustki va pastki qismlaridagi temperaturalarni bir xil deb hisoblang.

Berilgan	Formula	Yechilishi
$V_2 = 3.5V_1$ $T_1 = T_2$ h-?	$h = \frac{2.5P_2}{\rho g}$	$P_1V_1 = P_2V_2$ $(P_2 + \rho g h)V_1 = 3.5V_1P_2$ $\rho g h V_1 = 3.5V_1P_2 - V_1P_2$ $h = \frac{2.5P_2}{\rho g}$ $[h] = \frac{\frac{N}{m^2}}{\frac{kg}{m^3}\frac{N}{kg}} = m$ $h = \frac{2.5 * 100\,000}{1000 * 10} m = 25m$

6*. Berk idishdagi gazni 120 K ga qizdirilganda uning bosimi ikki marta ortgan bo‘lsa, gazning dastlabki temperaturasi qanday bo‘lgan?

Berilgan	Formula	Yechilishi
$V = const$ $\Delta T = 120K$ $P_2 = 2P_1$ $T_1 - ?$	$T_1 = T_2 - \Delta T$ $T_2 = 2\Delta T$	$\frac{p_1}{p_2} = \frac{T_1}{T_2}$; $\Delta T = T_2 - T_1$; $T_1 = T_2 - \Delta T$; $P_1T_2 = 2P_1T_1;\ T_2 = 2T_1;\ T_2 = 2T_2 - 2\Delta T$ $2\Delta T = 2T_2 - T_2$ $T_2 = 2\Delta T = 2 * 120K = 240K$ $T_1 = T_2 - \Delta T = 240K - 120K = 120K$

7*. Gaz izobarik ravishda temperaturasini 10 K ga oshirilganda, gaz hajmi dastlabki qiymatining 1/20 qismi qadar oshdi. Gazning dastlabki temperaturasi qanday bo‘lgan?

Berilgan	Formula	Yechilishi
$p = const$ $\Delta T = 10\ K$ $V_2 = 1.05V_1$ $T_1 - ?$	$T_1 = \frac{\Delta T}{0.05}$	$\Delta T = T_2 - T_1$ $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $\frac{V_1}{T_1} = \frac{1.05V_1}{\Delta T + T_1}$ $V_1\Delta T + V_1T_1 = 1.05V_1T_1;\ V_1\Delta T = 1.05V_1T_1 - V_1T_1$ $\Delta T = 0.05T_1$ $T_1 = \frac{\Delta T}{0.05} = \frac{10}{0.05} = 200K$

8*. Massasi 3 kg bo‘lgan ideal gaz 127 °C temperaturada erkin siljiydigan porshen ostida 2,5 m^3 hajmni egallab turibdi. Qanday temperaturada porshen ostidagi gazning zichligi 2 kg/m^3 bo‘ladi?

Berilgan	Formula	Yechilishi
$m = 3kg$ $t_1 = 127℃$ $V_1 = 2.5m^3$ $\rho_2 = 2\frac{kg}{m^3}$ $P = const$ $T_2 - ?$	$T_2 = \frac{T_1 m}{V_1 \rho}$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $T_2 = \frac{T_1 V_2}{V_1} = \frac{T_1 \frac{m}{\rho}}{V_1} = \frac{T_1 m}{V_1 \rho}$ $[T] = \frac{K * kg}{m^3 * kg/m^3} = K$ $T_2 = \frac{400 * 3}{2 * 2.5} K = 240\ K$

1. Temperaturasi 47 °C va ichki energiyasi 80 kJ bo‘lgan argon gazining massasini aniqlang.

Berilgan	Formula	Yechilishi
$\boldsymbol{t = 47℃}$ $\boldsymbol{U = 80\ kJ}$ $\boldsymbol{M = 40 * 10^{-3}\frac{kg}{mol}}$ $\boldsymbol{m - ?}$	$\boldsymbol{U = \frac{3}{2}\frac{m}{M}RT}$ $\boldsymbol{dan}$ $\boldsymbol{m = \frac{2MU}{3RT}}$ $\boldsymbol{[m] = \frac{\frac{kg}{mol}J}{\frac{J}{mol * K}K} = kg}$	$\boldsymbol{U = \frac{2 * 40 * 10^{-3} * 8 * 10^4}{8.31 * 320}}$ $\boldsymbol{= 0.24 * 10}$ $\boldsymbol{= 2.4\ kg}$

2. Bir atomli ideal gazning hajmi 0,4 m3 va ichki energiyasi 45 kJ bo‘lsa, uning bosimi nimaga teng?

Berilgan	Formula	Yechilishi
$\boldsymbol{V = 0.4\ m^3}$ $\boldsymbol{U = 45\ kJ}$ $\boldsymbol{p - ?}$	$\boldsymbol{U = \frac{3}{2}pV}$ $\boldsymbol{dan}$ $\boldsymbol{p = \frac{2}{3}\frac{U}{V}}$ $\boldsymbol{[p] = \frac{J}{m^3} = \frac{N * m}{m^3} = \frac{N}{m^2}}$ $\boldsymbol{= Pa}$	$\boldsymbol{p = \frac{2 * 45 * 10^3}{3 * 0.4}}$ $\boldsymbol{= 75 * 10^3\ Pa}$ $\boldsymbol{= 75\ kPa}$

3. Miqdori 3 mol neon gazi 40 °C dan – 80 °C gacha sovitilganda, uning ichki energiyasi qanday o‘zgaradi?

Berilgan	Formula	Yechilishi
$\boldsymbol{\nu = 3\ mol}$ $\boldsymbol{t_1 = 40℃}$ $\boldsymbol{t_2 = -80℃}$ $\boldsymbol{M = 20\frac{kg}{mol}}$ $\boldsymbol{\Delta U - ?}$	$\boldsymbol{\Delta U = \frac{3}{2}\frac{m}{M}RT}$ $\boldsymbol{\nu = \frac{m}{M}}$ $\boldsymbol{\Delta T = t_2 - t_1}$ $\boldsymbol{[U] = mol\frac{J}{mol * K}K = J}$	$\boldsymbol{\Delta U = \frac{3 * 3}{2}8.31 * 120}$ $\boldsymbol{= 4487.4J}$

4. Massasi 80 g bo‘lgan geliy gazi 20 °C dan 70 °C gacha qizdirilganda, uning ichki energiyasi qanday o‘zgaradi?

Berilgan	Formula	Yechilishi
$\boldsymbol{m = 80\ g}$ $\boldsymbol{t_1 = 20℃}$ $\boldsymbol{t_2 = 70℃}$	$\boldsymbol{\Delta U = \frac{3}{2}\frac{m}{M}RT}$	$\boldsymbol{\Delta U = \frac{3 * 3}{2}8.31 * 120}$ $\boldsymbol{= 4487.4J}$

$M = 2\frac{kg}{mol}$ $\Delta U - ?$	$\nu = \frac{m}{M}$ $\Delta T = t_2 - t_1$ $[U] = mol\frac{J}{mol * K}K = J$	

5. Idishdagi $4 \cdot 10^{25}$ ta molekulaga ega bo‘lgan bir atomli ideal gazning temperaturasi 72 K ga ortganda, uning ichki energiyasi qanday o‘zgaradi?

Berilgan	Formula	Yechilishi
$N = 4 * 10^{25}$ $\Delta t = 72\ K$ $\Delta U - ?$	$\Delta U = \frac{3}{2}\nu RT$ $\nu = \frac{N}{N_A}$ $[U] = mol\frac{J}{mol * K}K = J$	$\Delta U = \frac{3}{2}\frac{4 * 10^{25}}{6.02 * 10^{23}}8.31 * 72$ $= 59.6 * 10^3 J$ $= 59.6\ kJ$

6. 10^5 Pa bosim ostida turgan bir atomli ideal gazning hajmi izobarik ravishda 300 cm3 dan 500 cm3 gacha oshdi. Bunda gazning ichki energiyasi qanchaga o‘zgargan?

Berilgan	Formula	Yechilishi
$p = 10^5\ Pa$ $V_1 = 300 sm^3$ $V_2 = 500 sm^3$ $\Delta U - ?$	$U_1 = \frac{3}{2}pV_1$ $U_2 = \frac{3}{2}pV_2$ $\Delta U = U_2 - U_1$ $\Delta U = \frac{3}{2}p(V_2 - V_1)$ $[U] = \frac{N}{m^2}m^3 = N * m = J$	$\Delta U = \frac{3}{2} * 10^5(500 * 10^{-6}$ $- 300 * 10^{-6})$ $= 30J$

7. Cho‘g‘lanma lampochka yonganda, ichidagi gazning temperaturasi 17 °C dan 307 °C gacha ko‘tarilsa, uning ichidagi gazning ichki energiyasi necha marta ortadi?

Berilgan	Formula	Yechilishi
$t_1 = 17℃$ $t_2 = 307℃$ $\frac{U_2}{U_1} - ?$	$\frac{U_2}{U_1} = \frac{\frac{3}{2}\nu RT_2}{\frac{3}{2}\nu RT_1} = \frac{T_2}{T_1}$	$\frac{U_2}{U_1} = \frac{580\ K}{290\ K} = 2$

M 13

1. Silindrik idishdagi 160 kPa bosim ostida turgan gaz izobarik ravishda kengayib, 48 kJ ish bajardi. Bunda gazning hajmi qanchaga ortgan?

Berilgan	Formula	Yechilishi
$p = 160\ kPa$ $p = const$ $A = 48\ kJ$ $\Delta V - ?$	$A = p\Delta V$ $\Delta V = \frac{A}{p}$ $[V] = \frac{J}{Pa} = \frac{N * m}{N/m^2} = m^3$	$\Delta V = \frac{48000}{160000}m^3 = 0.3m^3$

2. Porshen ostidagi 400 g massali havo izobarik qizdirildi. Havo tashqi kuchlar ustidan 8 kJ ish bajargan bo ‘lsa, u necha gradusga qizigan?

Berilgan	Formula	Yechilishi
$m = 400\ g$ $A = 8\ kJ$ $M = 32 \cdot 10^{-3}$ kg/mol $\Delta T - ?$	$A = \frac{m}{M}R\Delta T$ $\Delta T = \frac{AM}{mR}$ $[K] = \frac{J * \frac{kg}{mol}}{kg * \frac{J}{molK}} = K$	$\Delta T = \frac{8000 * 32 * 10^{-3}}{0.4kg * 8.31} = 77K$

3. 100 kPa bosim ostida turgan ideal gaz izobarik kengayib, hajmi 100 cm^3 dan 300 cm^3 gacha ortdi. Bunda gaz qanday ish bajargan?

Berilgan	Formula	Yechilishi
$p = 100\ kPa$ $p = const$ $V_1 = 100\ sm^3$ $V_2 = 300\ sm^3$ $A-?$	$A = p\Delta V$ $\Delta V = V_2 - V_1$ $[J] = Pa * m^3 = \frac{N}{m^2} * m^3 = N * m = J$	$A = 10^5 * (3 * 10^{-4} - 1 * 10^{-4})J = 20J$

4*. Ichki diametri 5 cm bo‘lgan silindrga gaz qamalgan. Silindr porsheniga 50 N tashqi kuch ta’sir etib, gaz hajmini 10 cm^3 ga kamaytirdi. Tashqi kuch olingandan keyin gaz kengayib, dastlabki holatiga qaytdi. Tashqi kuch olingandan keyin siqilgan gaz qancha ish bajargan?

Berilgan	Formula	Yechilishi
$d = 5sm$ $F = 50N$ $\Delta V = 10\ sm^3$ $A-?$	$A = p\Delta V$ $p = \frac{F}{S}$ $S = \frac{\pi d^2}{4}$ $A = p\Delta V = \frac{F}{S}\Delta V = \frac{F}{\frac{\pi d^2}{4}}\Delta V$ $[A] = \frac{N}{m^2}m^3 = N * m = J$	$A = \frac{4 * 50}{3.14 * (5 * 10^{-2})^2}10^{-5}J = 0.25J$

M 14

1. Massasi 0,5 kg va solishtirma issiqlik sig‘imi 450 J/(kg·K) bo‘lgan jism 10 ° C dan 310 ° C gacha qizdirilganda, qancha issiqlik miqdorini qabul qiladi?

Berilgan	Formula	Yechilishi
$m = 0.5\ kg$ $c = 450\frac{j}{kg * K}$ $t_1 = 10℃$ $t_2 = 310℃$ $Q-?$	$Q = cm(t_2 - t_1)$ $[Q] = \frac{j}{kg * K} * kg * K = J$	$Q = 450 * 0.5 * (310 - 10)J = 67500J = 67.5kJ$

2. Massasi 3 kg bo‘lgan jism 20 ° C dan 500 ° C gacha qiziganda, 1281,6 kJ issiqlik miqdori olgan bo‘lsa, bu jism qanday moddadan tayyorlangan?

Berilgan	Formula	Yechilishi
$m = 3\ kg$ $t_1 = 20℃$ $t_2 = 500℃$ $Q = 1281.6\ kJ$ c-?	$Q = cm(t_2 - t_1)$ $c = \frac{Q}{m(t_2 - t_1)}$ $[c] = \frac{j}{kg * K}$	$c = \frac{1281.6 * 10^3}{3 * (500 - 20)}\frac{j}{kg * K} = 890\frac{j}{kg * K}$ **Alyuminiy:** $c = 890\frac{j}{kg * K}$

3. Normal sharoitda temperaturasi 20 ° C va hajmi 1,5 l bo‘lgan suv qaynaguncha, qancha issiqlik miqdorini oladi?

Berilgan	Formula	Yechilishi
$V = 1.5\ l$ $c = 4200\frac{j}{kg * K}$ $t_1 = 20℃$ $t_2 = 100℃$	$Q = cm(t_2 - t_1) = c\rho V(t_2 - t_1)$ $[Q] = \frac{j}{kg * K} * \frac{kg}{m^3} * K = J$	$Q = 4200 * 1000 * 1.5 * 10^{-3}(100 - 20)J = 504000J = 504kJ$

$Q-?$		

4. Normal sharoitda qaynab turgan suv ichida mis va qo ‘rg ‘oshindan yasalgan jismlar bor edi. Ular suvdan olingan paytda har biri qanday issiqlik miqdoriga ega bo ‘ladi? Misdan yasalgan jismning massasi 200 g, qo ‘rg ‘oshindan yasalgan jismning massasi 150 g ga teng deb oling.

Berilgan	Formula	Yechilishi
$m_1 = m_2 = 0.2\ kg$ $c_1 = 390\frac{j}{kg*K}$ $c_2 = 130\frac{j}{kg*K}$ $t = 100℃$ $Q_1-?\ Q_2-?$	$Q_1 = c_1mT$ $Q_2 = c_1mT$ $[Q] = \frac{j}{kg*K}*kg*K = J$	$Q_1 = 390*0.2*100J$ $= 7800J$ $Q_2 = 130*0.2*100J$ $= 2600J$

M 15

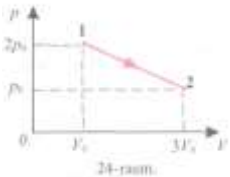
1. Temperaturasi 27 ₀C va ichki energiyasi 50 kJ bo ‘lgan geliy gazining massasi qancha?

Berilgan	Formula	Yechilishi
$t = 27℃$ T=300 K $U = 50\ kJ$ $M = 4\frac{kg}{mol}$ $m-?$	$U = \frac{3}{2}\frac{m}{M}RT$ dan $m = \frac{2MU}{3RT}$ $[m] = \frac{\frac{kg}{mol}J}{\frac{J}{mol*K}K} = kg$	$U = \frac{2*4*10^{-3}*5*10^4}{3*8.31*300} = 0.053*10$ $= 0.53\ kg$

2. Bir atomli gazning bosimi 30 % kamayib, hajmi 6 marta oshsa, uning ichki energiyasi qanday o ‘zgaradi?

Berilgan	Formula	Yechilishi
$p_2 = 0.7p_1$ $V_2 = 6V_1$ $\Delta U-?$	$U_1 = \frac{3}{2}p_1V_1$ $U_2 = \frac{3}{2}p_2V_2$ $\Delta U = U_2 - U_1$ $[U] = \frac{N}{m^2}m^3 = N*m = J$	$\Delta U = U_2 - U_1 = \frac{3}{2}p_2V_2 -$ $\frac{3}{2}p_1V_1 = \frac{3}{2}0.7p_16V_1 -$ $\frac{3}{2}p_1V_1 = 4.2U_1$ $\Delta U = 4.2U_1$

3. O ‘zgarmas massali bir atomli ideal gaz 1-holatdan 2-holatga o ‘tdi (24-rasm). Bunda gazning ichki energiyasi qanday o ‘zgaradi? Gazning Dastlabki bosimi p_0 =150 kPa va hajmi V_0 = 4 l bo ‘lgan.



Berilgan	Formula	Yechilishi
$p_1 = 150\ kPa$ $V_1 = 4\ l$ $p_2 = 0.5p_1$ $V_2 = 3V_1$ $\Delta U-?$	$U_0 = \frac{3}{2}p_0V_0$ $U_1 = \frac{3}{2}2p_0V_0$ $U_2 = \frac{3}{2}p_03V_0$	$U_0 = \frac{3}{2}*150*4 = 900J$ $U_1 = \frac{3}{2}*2*150*4 = 1800J$ $U_2 = \frac{3}{2}*150*3*4 = 2700J$ $\Delta U = 2700J - 1800J = 900J$

	$\Delta U = U_2 - U_1$ $[U] = \frac{N}{m^2}m^3 = N * m$ $= J$	

4. Silindr porsheni ostida massasi 1,6 kg massali kislorod gazi 17 ° C haroratda turibdi. Gaz izobarik kengayib 40 kJ ish bajargan bo ‘lsa, u qanday temperaturagacha qizigan?

$m = 1.6\ kg$ $A = 40\ kJ$ $t_1 = 17℃$ $T_1 = 290K$ $M = 32 \cdot 10^{3}$ kg/mol $\Delta T -?$	$A = \frac{m}{M}R\Delta T$ $\Delta T = \frac{AM}{mR}$ $T_2 = \Delta T + T_1$ $[K] = \frac{J * \frac{kg}{mol}}{kg * \frac{J}{molK}} = K$	$\Delta T = \frac{40000 * 32 * 10^{-3}}{1.6kg * 8.31}$ $= 96K$ $T_2 = 290 + 96 = 386K$ $t_2 = (386 - 273)℃ = 113℃$

5. Erkin siljiy oladigan porshen ostidagi temperaturasi 27 ℃, hajmi 10 l va bosimi 100 kPa bo ‘lgan ideal gaz 60 K ga izobarik qizdirildi. Bunda gaz tashqi kuchlar ustidan qanday ish bajaradi?

Berilgan	Formula	Yechilishi
$t_1 = 27℃$ $V_1 = 10\ l$ $p_1 = 100\ kPa$ $\Delta T = 60\ K$ $A -?$	**Gazning dastlabki hajmi bizga ma’lum. Gazning keying hajmini topamiz** $V_2 = \frac{T_2 V_1}{T_1}$ $A = p(V_2 - V_1)$ $[J] = Pa * m^3 = \frac{N}{m^2} * m^3$ $= N * m = J$	$V_2 = \frac{360 * 10^{-2} m^3}{300K}$ $= 1.2 * 10^{-2} m^3$

$$A = p(V_2 - V_1) = p\left(\frac{T_2 V_1}{T_1} - V_1\right) = pV_1\left(\frac{T_2}{T_1} - 1\right) = 10^5 * 10^{-2}\left(\frac{360}{300} - 1\right) = 0.2 * 10^3 = 200\,J$$

6. Modda miqdori 25 mol bo ‘lgan gaz 20 K ga isitilganda, izobarik kengayib uning hajmi dastlabki hajmiga nisbatan 20 % ga ortdi. Gazning dastlabki temperaturasi qanday bo ‘lgan? Gaz kengayishida bajarilgan ish nimaga teng?

Berilgan	Formula	Yechilishi
$\nu = 25\ mol$ $\Delta T = 20\ K$ $p = const$ $V_2 = 1.2V_1$ $T_1 -?\ \ A -?$	$V_2 = \frac{T_2 V_1}{T_1}$ $T_1 = \frac{T_2 V_1}{V_2} = \frac{(T_1 + \Delta T)V_1}{1.2V_1}$ $1.2T_1 = T_1 + \Delta T$ $0.2T_1 = \Delta T$ $T_1 = \frac{\Delta T}{0.2}$ $A = \nu R \Delta T$	$T_1 = \frac{20\ K}{0.2} = 100\ K$ $A = 25\ mol * 8.31\ \frac{J}{mol * K}$ $* 20K = 4165\ J$ $= 4.165\ kJ$

7. Massasi 8 kg va 90 ° C temperaturaga ega bo ‘lgan suvga 20 ° C temperaturadagi suvdan qancha qo ‘shganda, aralashma temperaturasi 30 ° C ga teng bo ‘ladi?

Berilgan	Formula	Yechilishi
$m_1 = 8\ kg$ $t_1 = 90$ ℃	$Q_1 = Q_2$	$m_2 = \frac{8\,(90 - 60)}{30 - 20} kg = 48kg$

$t_2 = 20$ ℃ $t_3 = 30$ ℃ m_2−?	$cm_1(t_1 - t_3) = cm_2(t_3 - t_2)$ $m_2 = \frac{m_1(t_1 - t_3)}{(t_3 - t_2)}$ $[m] = \frac{kg℃}{℃} = kg$	

8. Massasi va boshlang ‘ich temperaturasi bir xil bo ‘lgan vodorod va geliy gazlari izobarik ravishda 60 K ga qizdirildi. Vodorodni qizdirishda bajarilgan ishni va geliyni qizdirishda bajarilgan ish bilan taqqoslang.

Berilgan	Formula	Yechilishi
$\Delta T = 60\ K$ $m_1 = m_2$ $T_{01} = T_{02}$ $\frac{A_1}{A_2}$−?	$A_1 = \frac{m}{M_1}R\Delta T$ $A_2 = \frac{m}{M_2}R\Delta T$ $\frac{A_1}{A_2} = \frac{\frac{m}{M_1}R\Delta T}{\frac{m}{M_2}R\Delta T}$ $\frac{A_1}{A_2} = \frac{M_2}{M_1}$	$\frac{A_1}{A_2} = \frac{4 * 10^{-3}\ kg/mol}{2 * 10^{-3}\ kg/mol} = 2$ $A_1 = 2A_2$

9. 15 ° C temperaturadagi 125 l suv, 45 ° C temperaturali 25 l suv bilan aralashtirilsa, natijaviy temperatura qanday bo ‘ladi?

Berilgan	Formula	Yechilishi
$t_1 = 15$℃ $V_1 = 125\ l$ $t_2 = 45$ ℃ $V_2 = 25\ l$ t−?	$Q_1 = Q_2$ $cm_1(t - t_1) = cm_2(t_2 - t)$ $m = \rho V$ $t = \frac{\rho(V_2t_2 + V_1t_1)}{\rho(V_1 + V_2)}$ $= \frac{(V_2t_2 + V_1t_1)}{(V_1 + V_2)}$ $[t] = \frac{m^3 * ℃}{m^3} = ℃$	$t = \frac{25\ * 45 + 125 * 15}{125 + 25}$℃ $= 20$℃

10. Yuvinish vannasiga 10 ° C li sovuq suv va 90 ° C li issiq suv quyib, 50 ° C li iliq suv tayyorlandi. Agar vannadagi iliq suv hajmi 80 l bo ‘lsa, vannaga sovuq va issiq suvning har biridan qanchadan solingan? Vanna idishining olgan issiqlik miqdorini hisobga olmang.

Berilgan	Formula	Yechilishi
$t_1 = 10$℃ $t_2 = 90$ ℃ $t = 50$℃ $V = 80\ l$ V_1−? V_2−?	$Q_1 = Q_2$ $cm_1(t - t_1) = cm_2(t_2 - t)$ $m_1 + m_2 = m$ $V_1 + V_2 = V$ $V_1t - V_1t_1 = V_2t_2 - V_2t$ $V_1(t_2 - t_1) = V(t_2 - t)$ $V_1 = \frac{V(t_2 - t)}{t_2 - t_1}$ $V_2 = V - V_1$ $[V] = \frac{l * ℃}{℃} = l$	$V_1 = \frac{80 * 40}{80}l = 40\ l$ $V_2 = 80\ l - 40\ l = 40\ l$

11*. 800 m/s tezlik bilan uchib borayotgan po ‘lat o ‘q qumga tiqilib qoldi. O ‘qning urilishida ajralgan issiqlikning 54 % qumni isitishga ketsa, o ‘q necha gradusga isiydi? c_p = 460 J/(kg·K).

Berilgan	Formula	Yechilishi
$v = 800\frac{m}{s}$ $\eta = 0.54$ $Q_2 = 0.54Q_1$ $c = 460\,J/(kg*K)$ $\Delta t_1 - ?$	$Q = (1-\eta)E_K$ $mc\Delta t_1 = (1-\eta)\frac{mv^2}{2}$ $\Delta t_1 = \frac{(1-\eta)\frac{v^2}{2}}{c}$	$\Delta t_1 = \frac{(1-0.54)*(800)^2}{2*460} = 320℃$

$$[t] = \frac{(m/s)^2}{J/(kg*℃)} = \frac{(m/s)^2}{N*m/(kg*℃)} = \frac{(m/s)^2}{kg*\frac{m}{s^2}*m/(kg*℃)} = ℃$$



1. Massasi qanday bo ‘lgan spirt yonganda 5,8 MJ issiqlik miqdori ajralib chiqadi? Spirtning solishtirma yonish issiqligi 2,9·10$_7$ J/kg ga teng.

Berilgan	Formula	Yechilishi
$Q = 5.8\,MJ$ $q = 2.9*10^7\frac{J}{kg}$ $m - ?$	$Q = qm$ $m = \frac{Q}{q}$ $[m] = \frac{J}{J/kg} = kg$	$m = \frac{5.8*10^6}{2.9*10^7}kg = 0.2\,kg$

2. Massasi 25 kg bo ‘lgan toshko ‘mir batamom yonganda ajralib chiqadigan issiqlikni olish uchun, qancha quruq o ‘tin yoqish kerak bo ‘ladi?

Berilgan	Formula	Yechilishi
$m_1 = 2.5\,kg$ $q_1 = 29\frac{MJ}{kg}$ $q_2 = 10\frac{MJ}{kg}$ $Q_1 = Q_2$ $m_2 - ?$	$Q = q_1m_1$ $Q = q_2m_2$ $m_2 = \frac{q_1m_1}{q_2}$ $[m] = \frac{\frac{J}{kg}*kg}{\frac{J}{kg}} = kg$	$m_2 = \frac{29*10^6*2.5}{10*10^6}kg = 7.25\,kg$

3. Neksiya avtomashinasiga har yuz kilometrga o ‘rtacha 10 l benzin sarflansa, har bir kilometrda qancha issiqlik ajralib chiqadi? Benzinning zichligi 700 kg/m^3.

Berilgan	Formula	Yechilishi
$s = 100\,km$ $V = 10\,l$ $s_1 = 1\,km$ $\rho = 700\frac{kg}{m^3}$ $q = 46\frac{MJ}{kg}$ $Q_1 - ?$	**Proporsiya tuzamiz:** $100\,km$ __ $10\,l$ $1\,km$ ___ $V_1\,l$ $Q_1 = m_1q$ $m = \rho V$	$V_1 = \frac{1\,km*10\,l}{100\,km} = 0.1\,l = 0.1*10^{-3}m^3$ $m_1 = 700\frac{kg}{m^3}*0.1*10^{-3}m^3 = 0.07\,kg$ $Q_1 = 46\frac{MJ}{kg}*0.07\,kg = 3.22\,MJ$

4. O ‘choqda ovqat pishirish uchun 12 kg quruq o ‘tin yoqildi. O ‘tin yoqilganda ajralib chiqqan issiqlikning to ‘rtdan bir qismi ovqatga, qolgan

qismi o ‘choqni, qozonni va havoni isitishga ketdi. Ovqat pishguncha o ‘ziga qancha issiqlik miqdorini olgan?

Berilgan	Formula	Yechilishi
$m = 12\ kg$ $q = 10\frac{MJ}{kg}$ $\eta = 0.25$ $Q-?$	$Q = \eta mq$	$Q = 0.25 * 12\ kg * 10\frac{MJ}{kg}$ $= 30\ MJ$

1. Izotermik jarayonda gazga 5 kJ issiqlik berilgan bo ‘lsa, gaz ustida qancha
ish bajarilgan bo ‘ladi?

Berilgan	Formula	Yechilishi
$T = const$ $Q = 5\ kJ$ $A-?$	$Q = A$	$A = 5\ kJ$

2. Izoxorik jarayonda gazga 2,8 kJ issiqlik miqdori berilsa, gazning ichki energiyasi
qanchaga o ‘zgaradi?

Berilgan	Formula	Yechilishi
$V = const$ $Q = 2.8\ kJ$ $A-?$	$Q = \Delta U$	$\Delta U = 2.8\ kJ$

3. Gazga 3,5 kJ issiqlik berilganda uning ichki energiyasi 2,1 kJ ga ortadi. Gaz ustida qancha ish bajarilgan?

Berilgan	Formula	Yechilishi
$Q = 3.5\ kJ$ $\Delta U = 2.1\ kJ$ $A-?$	$Q = A + \Delta U$ $\Delta U = Q - A$	$\Delta U = 3.5\ kJ - 2.1\ kJ$ $= 1.4\ kJ$

4. Normal sharoitda bir atomli gazga issiqlik berilganda, gaz izobarik ravishda 0,05 m^3 kengaydi. Gazning ichki energiyasi qanday o ‘zgaradi?

Berilgan	Formula	Yechilishi
$p = const$ $p = 10^5\ Pa$ $\Delta V = 0.05\ m^3$ $\Delta U-?$	$Normal\ sharoit\ p$ $= 10^5\ Pa$ $\Delta U = \frac{3}{2}p\Delta V$	$U = \frac{3}{2}10^5\ Pa * 5 * 10^{-2}m^3$ $= 7.5 * 10^3 J$ $= 7.5\ kJ$

5. Metall ballondagi 25 mol bir atomli ideal gazning temperaturasini 20 K ga orttirish uchun unga qancha issiqlik miqdori berish kerak?

Berilgan	Formula	Yechilishi
$\nu = 25\ mol$ $\Delta T = 20\ K$ $Q-?$	$Q = A + \Delta U$ $V = const$ $bo'lganiuchun$ $Q = \Delta U = \frac{3}{2}\nu R\Delta T$	$Q = \frac{3}{2} * 25\ mol * 8.31\frac{J}{mol * K}20\ K =$ $= 6232.5\ J$

6. Silindrik idishdagi erkin siljiydigan porshen ostida bir atomli gaz bor. Gaz bosimi 10_5 Pa ga teng. Unga qancha issiqlik miqdori berilsa, hajmi 2 l ga oshadi?

Berilgan	Formula	Yechilishi
$p = const$ $p = 10^5\ Pa$ $\Delta V = 2\ l$ $Q-?$	$Q = A + \Delta U$ $A = \frac{m}{M}R\Delta T$ $\Delta U = \frac{3}{2}\frac{m}{M}R\Delta T$ $Q = p\Delta V + \frac{3}{2}p\Delta V = \frac{5}{2}p\Delta V$	$Q = \frac{5}{2}10^5 Pa * 2 * 10^{-3}m^3$ $= 500\ J$

7. Ballondagi bir atomli ideal gazga 500 J issiqlik miqdori berilganda, uning temperaturasi 40 K ga oshdi. Ballondagi gazning miqdori qanday bo ‘lgan?

Berilgan	Formula	Yechilishi
$Q = 500\ J$ $\Delta T = 40\ K$ $\nu-?$	$Q = A + \Delta U$ $V = const$ $bo'lganiuchun$ $Q = \Delta U = \frac{3}{2}\nu R\Delta T$ $\nu = \frac{2Q}{3R\Delta T}$	$\nu = \frac{2 * 500\ J}{3 * 8.31\frac{J}{mol * K}40\ K} = 1\ mol$



1. Ishchi jism (gaz) isitkichdan 840 J issiqlik oldi. Agar issiqlik dvigateli FIK 30 % bo ‘lsa, gaz qancha ish bajaradi?

Berilgan	Formula	Yechilishi
$Q = 840\ J$ $\eta = 30\ \%$ $A-?$	$\eta = \frac{A}{Q}$ $A = \eta Q$	$A = 0.3 * 840J = 252\ J$

2. Isitkichning temperaturasi 477 ° C, sovitkichniki 27 ° C bo ‘lgan issiqlik mashinasining maksimal FIKni hisoblang.

Berilgan	Formula	Yechilishi
$T_1 = 477\ ℃ = 750K$ $T_2 = 27\ ℃ = 300K$ $Q = 840\ J$ $\eta-?$	$\eta = \frac{T_1 - T_2}{T_1}$	$\eta = \frac{750K - 300K}{750K} = 0.6$ $= 60\%$

3. Ideal issiqlik dvigatelining FIK 62,5 % bo ‘lishi uchun uning isitkichdagi temperaturasi qanday bo ‘lishi kerak? Sovitkichning temperaturasi 300 K ga teng.

Berilgan	Formula	Yechilishi
$\eta = 62.5\%$ $T_2 = 300\ K$ $T_1 =?$	$\eta = \frac{T_1 - T_2}{T_1}$ $\eta T_1 = T_1 - T_2$ $T_2 = T_1 - \eta T_1$ $T_1 = \frac{T_2}{1 - \eta}$	$T_1 = \frac{300K}{1 - 0.625} = 800K$

4. Agar isitkichning temperaturasi 127 ° C, sovitkichning temperaturasi 7 ° C bo ‘lgan ideal issiqlik mashinasi bir siklda isitkichdan 1300 J issiqlik olsa, bajariladigan foydali ish nimaga teng?

Berilgan	Formula	Yechilishi

$T_1 = 400K$ $T_2 = 280K$ $Q_1 = 1300J$ $A-?$	$A = Q_1 - Q_2$ $\eta = \frac{A}{Q_1}$ $\eta = \frac{T_1 - T_2}{T_1}$	$\eta = \frac{400K - 280K}{400K} = 0.3$ $A = 0.3 * 1300J = 390J$

5. Foydali ish koeffitsiyenti 40 % bo ‘lgan issiqlik mashinasi bitta siklda sovitkichga 63 kJ issiqlik beradi. Mashina bitta siklda qancha ish bajaradi?

Berilgan	Formula	Yechilishi
$\eta = 0.4$ $Q_2 = 63kJ$ $A-?$	$A = Q_1 - Q_2;$ $\eta = \frac{Q_1 - Q_2}{Q_1} = \frac{A}{Q_1}$ $A = \frac{A}{\eta} - Q_2;$ $A = \frac{A - \eta Q_2}{\eta}$ $\eta A = A - \eta Q_2$ $\eta Q_2 = A - \eta A$ $A = \frac{\eta Q_2}{1 - \eta}$	$A = \frac{0.4 * 63000J}{1 - 0.4} =$ $= 42000J = 42kJ$

6. Ideal issiqlik dvigatelida sovitkichning temperaturasi 62 ° C, issiqlik dvigatelining FIK 50 % bo ‘lsa, isitkich bilan sovitkich temperaturalari orasidagi farq qanday?

Berilgan	Formula	Yechilishi
$\eta = 0.5\%$ $T_2 = 335\,K$ $\Delta T = ?$	$\eta = \frac{T_1 - T_2}{T_1}$ $T_1 = \frac{T_2}{1 - \eta}$	$T_1 = \frac{335K}{1 - 0.5} = 670K = 397℃$ $\Delta T = 670K - 335K = 335K$

7*. Karno siklida ishlaydigan ideal mashinada isitkich va sovitkich temperaturalarining nisbati 5 ga teng. Agar bir siklda sovitkichga 180 kJ issiqlik berilgan bo ‘lsa, isitkichdan olingan issiqlik miqdorini aniqlang.

Berilgan	Formula	Yechilishi
$\frac{T_1}{T_2} = 5$ $Q_2 = 180kJ$ $Q_1-?$	$\eta = \frac{T_1 - T_2}{T_1}$ $\eta Q_1 = Q_1 - Q_2$ $Q_1 = \frac{Q_2}{1 - \eta}$	$\eta = \frac{5T_2 - T_2}{5T_2} = 0.8$ $Q_1 = \frac{180000J}{(1 - 0.8)} = 900kJ$

8*. Ideal issiqlik dvigateli isitkichining temperaturasi 327 ° C bo ‘lib sovitkichning temperaturasi 127 ° C ga teng. Shu ideal mashinaning FIK ni ikki marta oshirish uchun isitkichning temperaturasini qancha oshirish kerak bo ‘ladi?

Berilgan	Formula	Yechilishi
$T_1 = 600K$ $T_2 = 400K$ $\eta_2 = 2\eta_1$ $\Delta T-?$	$\eta_1 = \frac{T_1 - T_2}{T_1}$ $\eta_2 = \frac{T_1{}' - T_2}{T_1{}'}$	$\eta_1 = \frac{600K - 400K}{600K} = 0.33$ $T_1' = \frac{400K}{1 - (2 * 0.33)} = 1176.8K$

	$T_1' = \frac{T_2}{1-\eta_2}$ $\Delta T = T_1' - T_1$	$\Delta T = T_1' - T_1 = 1176.8K - 600K$ $= 576.8K$

M19

1. O ‘choqda 60 kg po ‘latni 1400 ° C ga isitish uchun 4,6 kg maxsus yoqilg ‘i sarf bo ‘ladi. Agar po ‘latning solishtirma issiqlik sig ‘imi 460 J/kg·K, maxsus yoqilg ‘ining yonish issiqligi 30 MJ/kg bo ‘lsa, o ‘choqning issiqlik berishi (FIK) qanday?

Berilgan	Formula	Yechilishi
$m = 60kg$ $\Delta t = 1400℃$ $m_y = 4.6kg$ $c = 460\frac{J}{kg*K}$ $q = 30\frac{MJ}{kg}$ $\eta -?$	$Q_1 = qm_y$ $Q_2 = cm\Delta T$ $\eta = \frac{Q_1 - Q_2}{Q_1}$	$Q_1 = 30*10^6\frac{J}{kg}*4.6kg = 138*10^6J$ $Q_2 = 460\frac{J}{kg*K}*60kg*1400K$ $= 38.64*10^6J$ $\eta = \frac{138*10^6J - 38.64*10^6J}{138*10^6J} = 0.72$ $= 72\%$ O’ choq issiqlikning 28% ni o’ zida olib qoladi

2. Minutiga 4 g kerosin sarflaydigan isitkichda temperaturasi 31 ° C bo ‘lgan 2 l suv qancha vaqtdan so ‘ng qaynagan? Qurilmaning FIK 35 % $q_{kerosin}$ = 46 MJ/kg ga teng deb oling.

Berilgan	Formula	Yechilishi
$\tau = 1\ min$ $m_\tau = 4g$ $T_1 = 304K$ $V = 2l$ $T_2 = 373K$ $\eta = 0.35$ $q_{keros} = \frac{46MJ}{kg}$ $t -?$	$Q_1 = qm_{keros}$ $A_f = mc\Delta t = \rho V\Delta T$ $\eta = \frac{A_f}{Q}$	$A_f = 4200\frac{J}{kgK}*1000\frac{kg}{m^3}*2*10^{-3}m^3$ $*69K = 579600J$ $Q = \frac{579600J}{0,35} = 1656000J$ $m_k = \frac{Q_1}{q} = \frac{1.656*10^6J}{46*10^6\frac{J}{kg}} = 0.036kg = 36g$ $t = \frac{1min*36g}{4g} = 9\ min$

3. 72 km/h tezlikda harakatlanayotgan avtomobilning 2 km yo ‘ldagi benzin sarfini hisoblang. Avtomobilning quvvati 23 kW, FIK 25 % ga teng. Benzinning solishtirma yonish issiqligi 46 MJ/kg.

Berilgan	Formula	Yechilishi
$v = 72\ km/h$ $s = 2km$ $P = 23\ kW$ $\eta = 0.25$ $q = 46\frac{MJ}{kg}$ $V -?$	$t = \frac{s}{v}$ $A_f = Pt$ $Q = \frac{A_f}{\eta}$ $m = \frac{Q}{q}$ $V = \frac{m}{\rho}$	$t = \frac{2000m}{20\frac{m}{s}} = 100s$ $A_f = 23000W*100s = 2.3*10^6J$ $Q = \frac{2.3*10^6J}{0.25} = 9.2*10^6J$ $m = \frac{9.2*10^6J}{46*10^6J} = 0.2kg$ $V = \frac{0.2kg}{700\ kg/m^3} = 2.86*10^{-4}m^3 = 0.286\ l$

4. Agar quvvati 50 kW bo ‘lgan dizel dvigatelining foydali ish koeffitsiyenti 34 % bo ‘lsa, u uch soatda qancha yoqilg ‘i sarflaydi? Dizel yoqilg ‘isining solishtirma yonish issiqligi 42 MJ/kg ga teng.

Berilgan	Formula	Yechilishi
$P = 50\ kW$ $\eta = 0.34$ $t = 3\ h$ $q = 42\frac{MJ}{kg}$ $m-?$	$m = \frac{Q}{q}$ $Q = \frac{A_f}{\eta}$ $A_f = Pt$	$A_f = 50 * 10^3 W * 10800 s = 540 * 10^6 J$ $Q = \frac{540 * 10^6 J}{0.34} = 1588 * 10^6 J$ $m = \frac{1588 * 10^6 J}{42 * 10^6 \frac{J}{kg}} = 37.8 kg$

5. Ideal issiqlik mashinasidagi gaz isitkichdan olgan issiqligining 60 % ni sovitkichga beradi. Agar isitkichning temperaturasi 227 ° C bo ‘lsa, sovitkichning temperaturasi qanday bo ‘lgan?

Berilgan	Formula	Yechilishi
$\eta = 0.4$ $T_1 = 500K$ $T_2-?$	$\eta = \frac{T_1 - T_2}{T_1}$ $T_2 = T_1(1 - \eta)$	$T_2 = 500K(1 - 0.4) = 300K$

6. Ideal issiqlik mashinasida isitkichining absolyut temperaturasi sovitkichning absolyut temperaturasidan uch marta yuqori. Isitkich gazga 30 kJ issiqlik miqdori berganda u qancha ish bajaradi?

Berilgan	Formula	Yechilishi
$T_1 = 3T_2$ $Q = 30kJ$ $A-?$	$\eta = \frac{T_1 - T_2}{T_1}$ $A_f = \eta Q$	$\eta = \frac{3T_2 - T_2}{3T_2} = 0.67$ $A_f = 0.67 * 30kJ = 20.1\ kJ$

7*. G orizontal yo ‘lda mototsiklning dvigateli 60 km/h tezlikda 3,5 kW quvvatga erishadi. Agar dvigatelning FIK 25 % bo ‘lsa, motoroller 3,6 l benzin sarflab, qancha yo ‘lni bosib o ‘tadi? Benzinning solishtirma yonish issiqligi 46 MJ/kg, zichligi 0,7 g/cm3.

Berilgan	Formula	Yechilishi
$v = 60\frac{km}{h}$ $P = 3.5\ kW$ $\eta = 0.25$ $V = 3.6\ l$ $q = 46\frac{MJ}{kg}$ $\rho = 0.7\frac{q}{sm^3}$ $s-?$	$s = vt$ $t = \frac{A}{P}$ $A = \eta Q$ $Q = qm$ $m = \rho V$	$m = 0.7 * 10^3 \frac{kg}{m^3} * 3.6 * 10^{-3} m^3 = 2.52\ kg$ $Q = 46 * 10^6 \frac{J}{kg} * 2.52 kg = 116 * 10^6 J$ $A = 0.25 * 116 * 10^6 J = 29 * 10^6 J$ $t = \frac{29 * 10^6 J}{3.5 * 10^3 W} = 8.3 * 10^3 s = 2.3\ h$ $s = 60\frac{km}{h} * 2.3\ h = 138\ km$

8*. O ‘zgarmas 108 km/h tezlik bilan harakatlanayotgan avtomobil 46 km yo ‘lda 5 kg benzin sarfladi. Benzinning solishtirma yonish issiqligi $46 \cdot 10_6$ J/kg va dvigatelning FIK 24 % bo ‘lsa, avtomobilning foydali quvvatini aniqlang.

Berilgan	Formula	Yechilishi
$v = 108\frac{km}{h}$ $s = 46\ km$	$P = \frac{A}{t}$	$Q = 46 * 10^6 \frac{J}{kg} * 5 kg = 230 * 10^6 J$ $A = 0.24 * 230 * 10^6 J = 55.2 * 10^6 J$

$m = 5\ kg$ $q = 46\frac{MJ}{kg}$ $\eta = 0.24$ $P-?$	$t = \frac{s}{v}$ $A = \eta Q$ $Q = qm$	$t = \frac{46000m}{30\frac{m}{s}} = 1533\ s$ $P = \frac{55.2 * 10^6 J}{1533\ s} = 36 * 10^3 W = 36kW$



1. Suv kapillyarda 14 mm ga ko ‘tarilgan bo ‘lsa, uning diametri qanday bo ‘lgan?

Berilgan:	Formula:	Yechilishi:
$h = 14\ mm$ $\sigma = 0.073\ N/m$ $d-?$	$F = \sigma l$ $mg = \sigma l$ $\rho Vg = \sigma 2\pi r$ $\rho Shg = \sigma \pi d$ $\rho \frac{\pi d^2}{4} hg = \sigma \pi d$ $d = \frac{4\sigma}{\rho hg}$	$d = \frac{4 * 0.073\ N/m}{1000\frac{kg}{m^3} * 14 * 10^{-3}m * 10\frac{N}{kg}} =$ $= 20.84 * 10^{-4}m = 2.084mm$

2. Kapillyarda spirt 22 mm balandlikka ko ‘tarildi. Kapillyar radiusi qanday bo ‘lgan? Spirtning zichligi 800 kg/m³.

Berilgan:	Formula:	Yechilishi:
$h = 22\ mm$ $\sigma = 0.022\ N/m$ $r-?$	$F = \sigma l$ $mg = \sigma l$ $\rho Vg = \sigma 2\pi r$ $\rho Shg = \sigma 2\pi r$ $\rho \pi r^2 hg = \sigma 2\pi r$ $r = \frac{2\sigma}{\rho hg}$	$r = \frac{2 * 0.022\ N/m}{800\frac{kg}{m^3} * 22 * 10^{-3}m * 10\frac{N}{kg}} =$ $= 0.25 * 10^{-3}m = 0.25\ mm$

3. Radiusi 0,6 mm bo ‘lgan kapillyarda kerosin qanday balandlikka ko ‘tariladi? Kerosinning zichligi 800 kg/m³.

Berilgan:	Formula:	Yechilishi:
$r = 0.6\ mm$ $\sigma = 0.024\ N/m$ $r-?$	$F = \sigma l$ $mg = \sigma l$ $\rho Vg = \sigma 2\pi r$ $\rho Shg = \sigma 2\pi r$ $\rho \pi r^2 hg = \sigma 2\pi r$ $h = \frac{2\sigma}{\rho rg}$	$r = \frac{2 * 0.024\ N/m}{800\frac{kg}{m^3} * 0.6 * 10^{-3}m * 10\frac{N}{kg}} =$ $= 0.01\ m = 10\ mm$

4. Teshigining diametri 2 mm bo ‘lgan tomizg ‘ichdan tomadigan suv tomchisining massasini aniqlang.

Berilgan:	Formula:	Yechilishi:
$d = 2\ mm$ $\sigma = 0.073\ N/m$ $m-?$	$F = \sigma l$ $mg = \sigma l$ $mg = \sigma \pi d$ $m = \frac{\sigma \pi d}{g}$	$m = \frac{0.073\frac{N}{m} * 3.14 * 2 * 10^{-3}m}{10\frac{N}{kg}}$ $= 0.46 * 10^{-4}kg = 0.046g$ $= 46mg$

5. Ichki diametri 2 mm bo ‘lgan tomizg ‘ichdan uzilayotgan suyuqlik tomchisining massasi 15 mg ekanligini bilgan holda, shu suyuqlikning sirt taranglik koeffitsiyentini toping.

Berilgan:	Formula:	Yechilishi:
$d = 2\,mm$ $m = 15\,mg$ σ =?	$F = \sigma l$ $mg = \sigma l$ $\rho V g = \sigma \pi d$ $\sigma = \frac{mg}{\pi d}$	$\sigma = \frac{15 * 10^{-6} kg * 10 \frac{N}{kg}}{3.14 * 2 * 10^{-3} m} = 2.39 * 10^{-3} \frac{N}{m}$ $= 0.0239\ N/m$ Kerosin

6. Zichligi 0,9 g/cm3 bo ‘lgan suyuqlik diametri 1,5 mm bo ‘lgan kapillyar naydagi ko ‘tarilish balandligi 10 mm bo ‘lsa, shu suyuqlikning sirt taranglik koeffitsiyentini aniqlang.

Berilgan:	Formula:	Yechilishi:
$\rho = 0.9 \frac{g}{sm^3}$ $d = 1.5\,mm$ $h = 10\,mm$ σ =?	$F = \sigma l$ $mg = \sigma l$ $\rho \frac{\pi d^2}{4} hg = \sigma \pi d$ $\sigma = \frac{\rho d h g}{4\pi}$	σ $= \frac{0.9 * \frac{10^3 kg}{m^3} * 1.5 * 10^{-3} m * 10^{-2} m * 10 \frac{N}{kg}}{4 * 3.14}$ $= 10{,}8 * 10^{-6} \frac{N}{m} = 10.8\,mN/m$

7. Teshigining diametri 3 mm bo ‘lgan tomizg ‘ichda 20 cm3 suv bor. Uning sirt taranglik koeffitsiyenti 73 mN/m. Tomizg ‘ichdan hammasi bo ‘lib nechta tomchi tomadi?

Berilgan:	Formula:	Yechilishi:
$d = 3\,mm\ V = 20 sm^3\ \sigma = 73\,mN/m$ N =?	$m = \rho V$ $m_0 = \frac{\sigma \pi d}{g}$ $N = \frac{m}{m_0} = \frac{\rho V g}{\sigma \pi d}$	$N = \frac{1000 \frac{kg}{m^3} * 20 * 10^{-6} m^3 * 10 \frac{N}{kg}}{73 * 10^{-3} \frac{N}{m} * 3.14 * 3 * 10^{-3} m}$ $= 290.8\,ta$

8. Yer sirtidagi kapillyarda suv 15 mm ga ko ‘tariladi. Agar Oyda erkin tushish tezlanishi yerdagidan 6 marta kichik ekanligi ma’ lum bo ‘lsa, Oyda shu kapillyarda suv qancha balandlikka ko ‘tariladi?

Berilgan:	Formula:	Yechilishi:
$h_0 = 15\,mm$ $g = \frac{g_0}{6}$ h–?	$h_0 = \frac{2\sigma}{\rho r g_0}$ $h = \frac{2\sigma 6}{\rho r g} = 6h_0$	$h = 6 * 15mm = 90mm$

9. Sovun pufakchasining sirt yuzi 12 cm2 ortganda sirt energiyasi qanchaga o ‘zgaradi?

Berilgan:	Formula:	Yechilishi:
$\Delta S = 12 sm^2$ ΔW–?	$W_1 = 2\sigma S_1$ $W_2 = 2\sigma S_2$ $\Delta W = W_2 - W_1$ $\Delta W = 2\sigma \Delta S$	$\Delta W = 2 * 0.04 \frac{N}{m} * 12 * 10^{-4} m^2$ $= 96 * 10^{-6} J = 96\,\mu J$

10. Sovun pufakchasining radiusi 2 cm dan 3cm gacha kattalashganda sirt energiyasi qanday o ‘zgaradi?

Berilgan:
$R_1 = 2\ \text{cm} = 2 \cdot 10^{-2}\ \text{m}$
$R_2 = 3\ \text{cm} = 3 \cdot 10^{-2}\ \text{m}$
$\sigma = 4 \cdot 10^{-2}\ \text{N/m}$.

Topish kerak:

$\Delta W = ?$

Formulasi:

$W = 2\,\sigma\,S;$
$S = 4\pi R^2;$

$\Delta W = 2\sigma S_2 - 2\sigma S_1 =$
$= 2\sigma \cdot 4\pi (R_2^2 - R_1^2)$

$[\Delta W] = \frac{\text{N}}{\text{m}} \cdot \text{m}^2 = \text{N} \cdot \text{m} = \text{J}.$

Hisoblash:

$W = 2 \cdot 4 \cdot 10^{-2} \cdot 4 \cdot 3{,}14 \cdot$
$\cdot (9 \cdot 10^{-4} - 4 \cdot 10^{-4}) = 5 \cdot 10^{-4}\ \text{J}$

Javob: $\Delta W = 5 \cdot 10^{-4}$ J ga ortadi.

11. Nima sababdan qo ‘ldagi moy qoldiqlarini suv bilan yuvish qiyin, lekin kerosin bilan oson?

№	Suyuqliklar	σ, N/m	№	Suyuqliklar	σ, N/m
1	Simob	0,47	4	O‘simlik yog‘i	0,033
2	Suv	0,073	5	Kerosin	0,024
3	Sovunli etitma	0,04	6	Etil spirti	0,022

Kerosinning sirt taranglik koeffitsiyenti kichik bo’ lgani uchun

12. Samovar jo ‘mragidan tomayotgan suv tomchisi sovuq holda og ‘ir bo ‘ladimi yoki issiq holda?

Temperatura ortganda sirt taranglik koeffitsiyenti kamayadi,

$mg = \sigma l$ bo’ lgani uchun issiq suv tomchisining og’ irligi kamayadi

13. Nima uchun xamir isitkichda yumshamaydi balki qotadi?

Isitilganda kapillyarlik hodisasi ortadi, xamir tarkibidan suv ko’ proq chiqib ketadi, xamir qotadi

14. Sovuq suvning molekulalari issiq va qaynoq suvning molekulalaridan farq qiladimi? Muz molekulalaridan-chi?

Molekulalari farq qilmaydi, ularning harakatlaari farq qiladi

1. Diametri 2 cm bo ‘lgan po ‘lat arqonga og ‘irligi 30 kN bo ‘lgan yuk osilgan. Arqondagi mexanik kuchlanishni aniqlang.

Berilgan:	Formula:	Yechilishi:
$d = 2\ sm$ $P = 30\ kN$ σ–?	$\sigma = \frac{F}{S} = \frac{F}{\frac{\pi d^2}{4}}$	$\sigma = \frac{4 * 3 * 10^4 N}{3.14 * (2 * 10^{-2})^2} = 0.96 * 10^8 \frac{N}{m^2}$ $= 96 * 10^6\ Pa$

2. 18 kN cho ‘zish kuchi berilganda, $6 \cdot 10_7$ N/m2 mexanik kuchlanish hosil bo ‘lishi uchun po ‘lat sterjenning ko ‘ndalang kesim yuzasi qancha bo ‘lishi kerak?

Berilgan:	Formula:	Yechilishi:
$F = 18\ kN$ $\sigma = 6 * 10^7 \frac{N}{m^2}$ S–?	$S = \frac{F}{\sigma}$	$S = \frac{18 * 10^3 N}{6 * 10^7 \frac{N}{m^2}} = 3 * 10^{-4} m^2 = 3\ sm^2$

3. Mustahkamlik chegarasi 0,5 MPa va zichligi 4000 kg/m3 bo ‘lgan g ‘isht devorning balandligi eng ko ‘pi bilan qancha bo ‘lishi mumkin?

Berilgan:	Formula:	Yechilishi:

$\sigma = 0.5\ MPa$ $\rho = 4000\frac{kg}{m^3}$ $h-?$	$\sigma = \frac{F}{S} = \frac{mg}{\frac{V}{h}}$ $= \frac{mgh}{\frac{m}{\rho}} = gh\rho$ $h = \frac{\sigma}{g\rho}$	$h = \frac{0.5 * 10^6 \frac{N}{m^2}}{10\frac{N}{kg} * 4 * 10^3 \frac{kg}{m^3}} = 12.5\ m$

4. Uzunligi 80 cm va ko ‘ndalang kesim yuzi 0,5 mm^2 bo ‘lgan simga massasi 25 kg bo ‘lgan yuk osilganda sim 2 mm ga uzaydi. Shu sim uchun Yung modulini aniqlang.

Berilgan:	Formula:	Yechilishi:
$l = 80\ sm$ $S = 0.5\ mm^2$ $m = 25\ kg$ $\Delta l = 2\ mm$ $E-?$	$\sigma = E\varepsilon = E\frac{\Delta l}{l}$ $\frac{F}{S} = E\frac{\Delta l}{l}$ $E = \frac{mgl}{S\Delta l}$	$E = \frac{25\ kg * 10\frac{N}{kg} * 0.8\ m}{0.5 * 10^{-6}m^2 * 2 * 10^{-3}m} = 200 * 10^9\ Pa$ $= 200\ GPa$

5. Po ‘latdan yasalgan sterjen uchiga 7,85 kN kuch qo ‘yilganda u uzilib ketdi. Uning diametri qanday bo ‘lgan? Po ‘lat uchun mustahkamlik chegarasi 170 MPa.

Berilgan:	Formula:	Yechilishi:
$F = 7.85\ kN$ $\sigma = 170\ MPa$ $d-?$	$\sigma = \frac{F}{S} = \frac{F}{\frac{\pi d^2}{4}}$ $d = \sqrt{\frac{4F}{\sigma\pi}}$	$d = \sqrt{\frac{4 * 7.85 * 10^3 N}{170 * 10^6 \frac{N}{m^2} * 3.14}} = \sqrt{60 * 10^{-6}m^2}$ $= 7.74 * 10^{-3}m = 7.74\ mm$

6*. Bir uchidan osib qo ‘yilgan po ‘lat sim suvga tushirilmoqda. Sim o ‘zining og ‘irligi ta’ sirida uzilib ketmasligi uchun simning uzunligi qancha bo ‘lishi kerak? Po ‘lat uchun mustahkamlik chegarasi 170 MPa, zichligi 7800 kg/m^3 ga teng.

Berilgan:	Formula:	Yechilishi:
$\sigma = 170\ MPa$ $\rho = 7800\frac{kg}{m^3}$ $l-?$	$\sigma = \frac{F - F_A}{S}$ $= \frac{\rho Vg - \rho_s Vg}{\frac{V}{l}}$ $l = \frac{\sigma}{g(\rho - \rho_s)}$	$l = \frac{170 * 10^6\ N/m^2}{10\frac{N}{kg}(1000 - 7800)kg/m^3} = 2500\ m$

1. Erish temperaturasida turgan 3 kg muzni suvga aylantirish uchun unga qancha issiqlik miqdori berish kerak?

Berilgan:	Formula:	Yechilishi:
$m = 3\ kg$ $\lambda = 334000\frac{J}{kg}$ $Q-?$	$Q = \lambda m$	$Q = 334000\frac{J}{kg} * 3\ kg = 1002000\ J = 1\ MJ$

2. Erish temperaturasida turgan m massali qalayni to ‘liq eritishga 10 kJ issiqlik miqdori sarflandi. Eritilgan qalayning massasini toping.

Berilgan:	Formula:	Yechilishi:
$\lambda = 60000\frac{J}{kg}$ $Q = 10\ kJ$ $m-?$	$Q = \lambda m$ $m = \frac{Q}{\lambda}$	$m = \frac{10000J}{60000\frac{J}{kg}} = 0.17\ kg$

3. Muzlatkichga qo ‘yilgan $0\ ^\circ$ C dagi $0,5\ l$ suv batamom muzlaguncha undan qancha issiqlik ajralib chiqadi?

Berilgan:	Formula:	Yechilishi:
$V = 0.5\ l$ $\lambda = 334000\frac{J}{kg}$ $Q-?$	$Q = \lambda m$ $Q = \lambda \rho V$	$Q = 334000\frac{J}{kg} * 1000\frac{kg}{m^3} * 0.5 * 10^{-3}\ m^3$ $= 167000J = 167\ kJ$

4. Erish temperaturasida turgan 5 kg jismni batamom eritguncha 420 kJ issiqlik miqdori sarflandi. Bu jism qaysi moddadan tayyor langan?

Berilgan:	Formula:	Yechilishi:
$m = 5\ kg$ $Q = 420\ kJ$ $\lambda-?$	$Q = \lambda m$ $\lambda = \frac{Q}{m}$	$\lambda = \frac{420000J}{5\ kg} = 64000\frac{J}{kg}$ $Oltin$

5. Temperaturasi $0\ ^\circ$ C bo ‘lgan $1\ l$ suvni qaynatish uchun sarflanadigan energiya shunday temperaturadagi qancha muzni eritishi mumkin?

Berilgan:	Formula:	Yechilishi:
$V = 1\ l$ $m-?$	$Q_1 = cm(t_2 - t_1)$ $Q_1 = \lambda m$ $c\rho V\Delta t = \lambda m$ $m = \frac{c\rho V\Delta t}{\lambda}$	$m = \frac{4200\frac{J}{kg*K} * 1000\frac{kg}{m^3} * 10^{-3}m^3 * 100K}{334000\frac{J}{kg}}$ $= 1.26\ kg$

6. Sirtining yuzasi 250 m^2 bo ‘lgan hovuz suvi $0\ ^\circ$ C temperaturada 1 mm qalinlikdagi
muz bilan qoplandi. Bunda atrofga qancha issiqlik miqdori ajralgan? Muzning zichligini 900 kg/m^3 ga teng deb oling.

Berilgan:	Formula:	Yechilishi:
$S = 250m^2$ $h = 1mm$ $\rho = 900\frac{kg}{m^3}$ $\lambda = 334000\frac{J}{kg}$ $Q-?$	$Q = \lambda m$ $Q = \lambda \rho V$ $Q = \lambda \rho Sh$	$Q = 334000\frac{J}{kg} * 900\frac{kg}{m^3} * 250\ m^2 * 10^{-3}m$ $= 75150000J = 75.15MJ$

1. Qaynash temperaturasida 5 kg suvni to ‘la bug ‘ga aylanishi uchun qancha issiqlik miqdori kerak? Suvning solishtirma bug ‘lanish issiqligi $2,3 \cdot 10^6$ J/kg.

Berilgan:	Formula:	Yechilishi:
$m = 5\ kg$ $r = 2.3 * 10^6\frac{J}{kg}$ $Q_b-?$	$Q_b = rm$	$Q_b = 2.3 * 10^6\frac{J}{kg} * 5kg = 11.5 * 10^6J$ $= 11.5\ MJ$

2. Massasi 50 g bo ‘lgan bug ‘ kondensatsiyalanganda qancha issiqlik ajraladi?

Berilgan:	Formula:	Yechilishi:
$m = 50\ g$ $r = 2.3 * 10^6 \frac{J}{kg}$ $Q_b - ?$	$Q_b = rm$	$Q_b = 2.3 * 10^6 \frac{J}{kg} * 0.05 kg = 115 * 10^3 J = 115\ kJ$

3. 20 ° C temperaturada 4 m³ havoda 40 g suv bug ‘i bo ‘lsa, havoning nisbiy namligi qancha? 20 ° C temperaturada to ‘yingan suv bug ‘ining zichligi 17,3 g/m³.

Berilgan:	Formula:	Yechilishi:
$V = 4m^3$ $m = 40\ g$ $\rho_0 = 17.3 \frac{g}{m^3}$ $\varphi - ?$	$\varphi = \frac{\rho}{\rho_0} 100\%$ $\rho = \frac{m}{V}$	$\rho = \frac{40\ g}{4m^3} = 10\ g/m^3$ $\varphi = \frac{10 \frac{g}{m^3}}{17.3 \frac{g}{m^3}} 100\% = 58\%$

4. 20 ° C temperaturada havodagi suv bug ‘ining bosimi 1,54 kPa ga teng. Agar 20 ° C temperaturada to ‘yingan suv bug ‘ining bosimi 2,43 kPa bo ‘lsa, nisbiy namlikni aniqlang.

Berilgan:	Formula:	Yechilishi:
$P = 1.54\ KPa$ $P_0 = 2.43\ kPa$ $\varphi - ?$	$\varphi = \frac{P}{P_0} 100\%$	$\varphi = \frac{1.54}{2.43} 100\ \% = 63\ \%$

5. 20 ° C temperaturada havodagi suv bug ‘ining zichligi 17 g/m³ ni tashkil etadi. Agar havoning absolyut namligi 11 g/m³ bo ‘lsa, nisbiy namligi qancha?

Berilgan:	Formula:	Yechilishi:
$\rho_0 = 17 \frac{g}{m^3}$ $\rho = 11 \frac{g}{m^3}$ $\varphi - ?$	$\varphi = \frac{\rho}{\rho_0} 100\%$	$\varphi = \frac{11}{17} 100\% = 65\ \%$

6. 24 ° C temperaturada havoning nisbiy namligi 50 % bo ‘lsa, absolyut namlik qanday bo ‘ladi? 24 ° C temperaturada to ‘yingan suv bug ‘ining zichligi 21,8 g/m³.

Berilgan:	Formula:	Yechilishi:
$\rho_0 = 21.8 \frac{g}{m^3}$ $\varphi = 50\ \%$ $\rho = ?$	$\varphi = \frac{\rho}{\rho_0} 100\%$ $\rho = \frac{\varphi \rho_0}{100\%}$	$\rho = \frac{50\ \% * 21.8 \frac{g}{m^3}}{100\ \%} = 10.9 \frac{g}{m^3}$

7. Psixrometrning quruq termometri 24 ° C ni, nam termometri 19 ° C ni ko ‘rsatmoqda. Psixrometrik jadvaldan foydalanib, havoning nisbiy namligini aniqlang.

Psixrometrik jadval

Quruq termometrning ko'rsatishi, °C	Quruq va nam termometrlar ko'rsatishlarining farqi, °C										
	0	1	2	3	4	5	6	7	8	9	10
15	100	90	80	71	61	52	44	36	27	20	12
16	100	90	81	71	62	54	46	37	30	22	15
17	100	90	81	72	64	55	47	39	32	24	17
18	100	91	82	73	65	56	49	41	34	27	20
19	100	91	82	74	65	58	50	43	35	29	22
20	100	91	83	74	66	59	51	44	37	30	24
21	100	91	83	75	67	60	52	46	39	32	26
22	100	92	83	75	68	61	54	47	40	34	28
23	100	92	84	76	69	61	55	48	42	36	30
24	100	92	84	77	69	62	56	49	43	37	31
25	100	92	84	77	70	63	57	50	44	38	33
26	100	92	85	78	71	64	58	51	46	40	34
27	100	92	85	78	71	65	59	52	47	41	36
28	100	93	85	78	72	65	59	53	48	42	29
	Nisbiy namlik, %										

Javobi: Psixrometrning quruq termometri 24 ° C, nam termometri 19 ° C ni ko ‘rsatmoqda. Bunda termometrlardagi farq 5 ° C ni tashkil etadi. Psixrometrik jadvaldan havoning nisbiy namligi 62 % ekanligini aniqlash mumkin.

M 24

1. Yerdan Quyoshgacha o‘rtacha masofa 149,6 mln km, Yupiterdan Quyoshgacha o‘rtacha masofa 778,3 mln km ga teng. Yer Quyosh bilan Yupiter oralig‘idagi holatda bo‘lsa, Yupiterdan qaytgan nur qancha vaqtda Yerga yetib keladi?

Berilgan:	**Formula:**	**Yechilishi:**
$s_1 = 149.6\ mln\ km$ $s_2 = 778.3\ mln\ km$ $s = \frac{s_2}{2}$ $t-?$	$t = \frac{s}{c}$	$t = \frac{\frac{778.3 * 10^9 m}{2}}{3 * 10^8 \frac{m}{s}} = 1297s$

2. Quyosh nuri Yerga qancha vaqtda yetib keladi? Yerdan Oygacha o‘rtacha masofa 384 ming km bo‘lsa, Oydan yorug‘lik nuri qancha vaqtda yetib keladi? Yerdan Quyoshgacha bo‘lgan masofa 149,6 mln km.

Berilgan:	**Formula:**	**Yechilishi:**
$s_1 = 149.6\ mln\ km$ $s_2 = 384000\ km$ $t_1-?$ $t_2-?$	$t_1 = \frac{s_1}{c}$ $t_2 = \frac{s_2}{c}$	$t_1 = \frac{149.6 * 10^9 m}{3 * 10^8 \frac{m}{s}} = 498.7s$ $t_2 = \frac{384 * 10^6 m}{3 * 10^8 \frac{m}{s}} = 1.28s$

3. Fizo usuli bilan yorug‘lik tezligini aniqlashda tishli g‘ildirak ko‘zgudan 8633 m masofaga joylashtirilgan. G‘ildirak 720 ta tishga ega. Tajribada yorug‘lik tezligi 313000 km/s bo‘lib chiqdi. G‘ildirak aylanish chastotasi qanday bo‘lgan?

Berilgan:	**Formula:**	**Yechilishi:**
$l = 8633\ m;\ N = 720$ $v = 313000 \frac{km}{s}$ $\nu-?$	$\nu = \frac{v}{4lN}$	$\nu = \frac{313 * 10^6 \frac{m}{s}}{4 * 8633m * 720} = 12.6s^{-1}$

M 25

Tushgan va qaytgan nurlar orasidagi burchak 70° bo‘lishi uchun yassi ko‘zguga nur qanday burchak ostida tushishi lozim?

Berilgan:	**Formula:**	**Yechilishi:**
$\alpha + \beta = 70°$ $\alpha-?$	$\alpha = \beta;\ \ \alpha = \frac{\alpha+\beta}{2}$	$\alpha = \frac{70°}{2} = 35°$

1. Yassi ko‘zguda buyum tasviri ko‘zgudan 60 cm masofada hosil bo‘lsa, buyum bilan uning tasviri orasidagi masofa qanday bo‘ladi?

Berilgan:	Formula:	Yechilishi:
$f = 60\ sm$ $d + f -?$	$d = f$ $d + f = f + f = 2f$	$d + f = 2 * 0.6m = 1.2m$

2. Absolyut sindirish ko‘rsatkichi 2 ga teng bo‘lgan muhitda yorug‘lik qanday tezlik bilan tarqaladi?

Berilgan:	Formula:	Yechilishi:
$n = 2$ $v -?$	$n = \frac{c}{v}\ \ v = \frac{c}{n}$	$v = \frac{3 * 10^8\ m/s}{2} = 1.5 * 10^8\ m/s$

3. Havodan shishaga tushgan va qaytgan nurlar orasidagi burchak 60° ga teng. Agar shishaning sindirish ko‘rsatkichi 1,5 ga teng bo‘lsa, sinish burchagi qanday bo‘ladi?

Berilgan:	Formula:	Yechilishi:
$\alpha + \gamma = 60°$ $n = 1.5\ ;\ \beta -?$	$\alpha = \gamma;\ \ n = \frac{sin\alpha}{sin\beta};\ \ sin\beta = \frac{sin\alpha}{n}$	$sin\beta = \frac{sin30°}{1.5} = \frac{0.5}{1.5} = 0.33$ $\beta = 19°$

Nur suvdan shishaga o‘tmoqda. Suvning sindirish ko‘rsatkichi 1,33 ga, shishaniki 1,5 ga teng. Bir xil vaqt ichida shu moddalardan yorug‘lik nuri o‘tgan masofalar nisbati qanday bo‘ladi?

$n_1 = 1.3;\ \ n_2 = 1.5;\ \ t_1 = t_2\ ;\ \frac{s_1}{s_2} -?$	$\frac{s_1}{s_2} = \frac{v_1}{v_2} = \frac{\frac{c}{n_1}}{\frac{c}{n_2}} = \frac{n_2}{n_1}$	$\frac{s_1}{s_2} = \frac{1.5}{1.3} = 1.15$